HAVE CONTROL ⟶ HAVE PATIENCE ⟶ HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PHYSICS

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. A boat is traveling due east at 12 m/s. A flag on the boat flaps at 53° N of W. Another flag on the shore flaps due north.

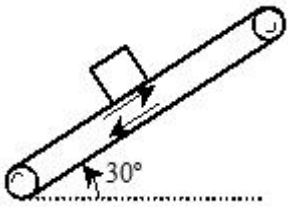
(A) Speed of wind with respect to ground is 16 m/s

(B) Speed of wind with respect to ground is 20 m/s

(C) Speed of wind with respect to boat is 20 m/s

(D) Speed of wind with respect to boat is 16 m/s

2. A block of mass 1 kg is stationary with respect to a conveyor belt that is accelerating with 1 m/s^2 upwards at an angle of 30° as shown in figure. What can be the coefficient of static friction between the block and the belt?



(A) 0.4 (B) 0.6 (C) 0.8 (D) 1.0

3. A ball of mass m collides elastically with an identical ball at rest with some impact parameter.

(A) 100 % energy transfer can never take place

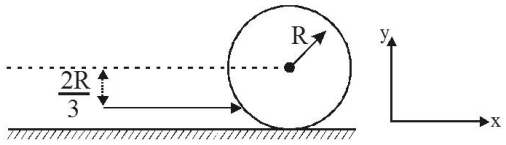
(B) 100 % energy transfer may take place

(C) angle of divergence between the two balls must be 90°

(D) angle of divergence between the two balls depend on impact parameter

Space for Rough Work

4. A billiard ball initially at rest is given a sharp blow by a cue stick. The impulse is horizontal and is applied at a distance 2R/3 below the centreline, as shown in figure. The initial speed of the ball is v_0, and the coefficient of kinetic friction is μ_k.



(A) Initially kinetic friction acts in $-\hat{i}$ direction

(B) Initially kinetic friction in $\hat{i}$ direction.

(C) Ball instantaneously starts pure rolling.

(D) Initial angular velocity of ball is $\frac{5v_0}{3R}$

5. Which of the following statement(s) is/are true ?

(A) Maximum velocity of photoelectrons depends on frequency and intensity of the incident light

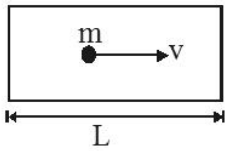
(B) Maximum velocity of photoelectrons depends only on the frequency of the incident light

(C) Photoelectric current increases with increase in intensity of incident light by increasing only the frequency.

(D) Photoelectric current increases with increase in intensity of incident light by increasing the number of photons.

Space for Rough Work

6. Figure shows a small particle in a box which bounces back and forth at constant speed. The reflections of particle from the wall creates a standing wave analogous to standing wave on a string tied at both ends. Since a standing wave confined to a region can have only selected wavelengths, momentum of the particle is quantized. We can safely assume that such a particle only has kinetic energy. This energy must also be quantized.



(A) The momentum of particle in n^{th} mode of standing wave is $\frac{nh}{2L}$

(B) Particle's energy in n^{th} state is $\frac{h^2n^2}{8mL^2}$

(C) Matter wave in this case will be electromagnetic in nature.

(D) None of these

7. The ^{238}U nucleus has a binding energy of about 7.6 MeV per nucleon. If the nucleus were to fission into two equal fragments, each would have a kinetic energy of just over 100 MeV. From this, it can be concluded that

(A) nuclei near A = 119 has mass less than half that of ^{238}U.

(B) nuclei near A = 119 have masses greater than half that of ^{238}U

(C) nuclei near A = 119 must be bound by about 6.7 MeV/ nucleon

(D) nuclei near A = 119 must be bound by about 8.5 MeV/ nucleon

Space for Rough Work

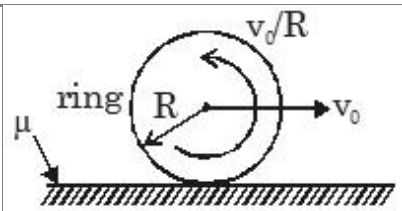
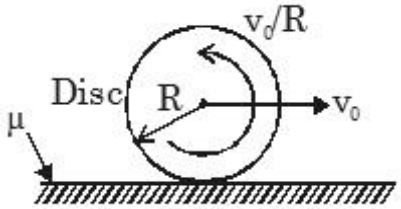
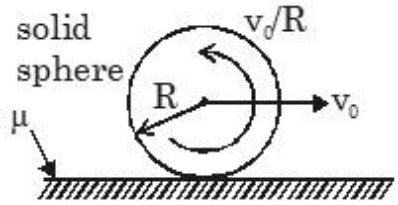
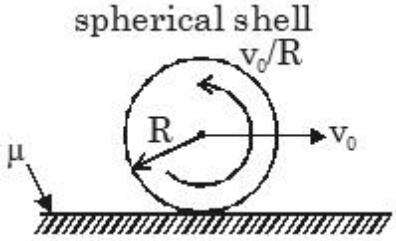
SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–1	In all other cases.

Space for Rough Work

Paragraph for Questions 8 - 10

Column–1		Column-2		Column-3	
Mass of the Object is 'm' situation is given at t = 0		Kinetic energy at the time when slipping ceases.		Time till slipping ceases.	
(I)		(i)	Zero	(P)	$\frac{v_0}{\mu g}$
(II)		(ii)	$\frac{1}{12}mv_0^2$	(Q)	$\frac{2v_0}{3\mu g}$
(III)		(iii)	$\frac{9}{70}mv_0^2$	(R)	$\frac{4v_0}{7\mu g}$
(IV)		(iv)	$\frac{mv_0^2}{30}$	(S)	$\frac{4v_0}{5\mu g}$

1. Name the object that has maximum loss in kinetic energy, its kinetic energy after slipping ceases and the time till slipping ceases.

(A) (II) (iii) (Q) (B) (I) (i) (P) (C) (III) (iii) (R) (D) (IV) (iv) (S)

Space for Rough Work

2. Name the object has minimum loss in kinetic energy during slipping, its kinetic energy after slipping ceases and the time till slipping ceases :-

(A) (II) (ii) (P) (B) (I) (i) (S) (C) (III) (iii) (R) (D) (IV) (iv) (Q)

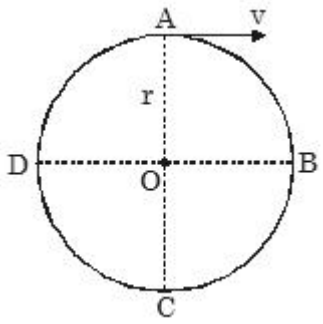
3. Name the object, which stops when slipping ceases, its kinetic energy at the time of slipping ceases and time till slipping ceases :-

(A) (II) (ii) (R) (B) (III) (iii) (S) (C) (IV) (iv) (Q) (D) (I) (i) (P)

Space for Rough Work

Paragraph for Questions 11 - 13

A particle is going around a circle with constant speed v of radius r. At t = 0 particle is at A. Particle takes 4 seconds to complete one round.



Column-1		Column-2		Column-3	
magnitude of displacement (in m)		Direction of displacement vector		Magnitude of change in velocity (in m/s)	
(I)	2R	(i)		(P)	$\sqrt{2}V$
(II)	$2\sqrt{2}R$	(ii)		(Q)	2V
(III)	$\sqrt{2}R$	(iii)		(R)	$2\sqrt{2}V$
(IV)	R	(iv)		(S)	V

4. Between A & B :-

(A) (II) (iv) (P) (B) (I) (iii) (S) (C) (III) (iv) (P) (D) (III) (i) (P)

Space for Rough Work

5. Between B & D :-

(A) (I) (iii) (R) (B) (I) (ii) (Q) (C) (II) (ii) (R) (D) (I) (ii) (P)

6. From t = 2 to t = 4 sec :-

(A) (I) (iii) (Q) (B) (I) (ii) (R) (C) (III) (iii) (Q) (D) (IV) (i) (P)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

MATCHING LIST TYPE QUESTION (ANSWER AS THREE DIGIT INTEGER)

This section contains **FIVE (05)** questions.

The answer to each question is a **THREE DIGIT INTEGER** ranging from 111 to 999, both inclusive.

Each question having two matching lists. Choices for the correct combination of elements form List-I (P), (Q), (R) and List-II (1), (2), (3).

Given matching type questions in which List-I (P,Q,R) match with List-II (1,2,3) and represent your answer as four digit integer for example correct match for P $\longrightarrow$ 2, Q $\longrightarrow$ 3, R $\longrightarrow$ 1.

Each matching in List-II may be/may not be correct & can be used more than one time for matching in **List-I**. For example your answer can be 223, 133, 143 etc...

For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

---

1. Two blocks of masses 3 kg and 6 kg are connected by an ideal relaxed spring and are placed on a frictionless horizontal surface. The 3 kg block is imparted a speed of 2 m/s towards left. (k = 2 N/m)



List-I		List-II	
(P)	The speed of 3 kg block is $\frac{2}{3}$ m/s at time	(1)	π sec
(Q)	The speed of 6 kg block is $\frac{4}{3}$ m/s at time	(2)	$\frac{\pi}{3}$ sec
(R)	The speed of 3 kg block is $\frac{4}{3}$ m/s at time	(3)	$\frac{\pi}{4}$ sec

---

Space for Rough Work

2. Three physical quantities are listed in list–I and their values are listed in list–II in random order.

List-I		List-II	
(P)	Binding energy of heavy nuclei per nucleon	(1)	10 keV
(Q)	X-ray photon energy	(2)	7 MeV
(R)	Photon energy of visible light	(3)	2 eV

3. Match the statements given in List-I with their corresponding possible results in List-II.

List-I		List-II	
(P)	If photons of ultraviolet light of energy 12 eV are incident on a metal surface of work function of 3eV, then the stopping potential (in eV) will be	(1)	8
(Q)	The ratio of wavelengths of K_α lines of two elements is $\left(\frac{85}{81}\right)^2$. Number of elements having atomic numbers between these elements will be	(2)	3
(R)	If 20 gm of a radioactive substance due to radioactive decay reduces to 10gm in 4 minutes, then in what time (in minutes) 80 gm of the same substance will reduce to 20 gm.	(3)	9

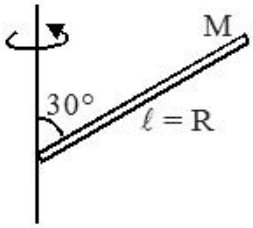
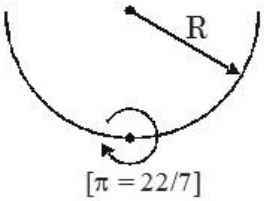
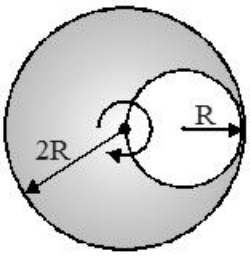
Space for Rough Work

4. Two particles A and B moving in x-y plane are at origin at t = 0 sec. The initial velocity vectors of A and B are $\vec{u}_A = 8\hat{i}$ m/s and $\vec{u}_B = 8\hat{j}$ m/s. The acceleration of A and B are constant and are $\vec{a}_A = -2\hat{i}$ m/s^2 and $\vec{a}_B = -2\hat{j}$ m/s^2. List-I gives certain statements regarding particle A and B. List-II gives corresponding results. Match the statements in List-I with corresponding results in List-II.

List-I		List-II	
(P)	The time (in seconds) at which velocity of A relative to B is zero	(1)	$16\sqrt{2}$
(Q)	The distance (in metres) between A and B when their relative velocity is zero	(2)	$8\sqrt{2}$
(R)	The magnitude of relative velocity of A and B at the instant they are at same position.	(3)	4

Space for Rough Work

5. Match the value of moment of inertia of structures in List-I with corresponding results in List-II.

List-I		List-II	
(P)	 Uniform Rod	(1)	$\frac{8MR^2}{11}$
(Q)	 Uniform Semicircular Ring. Axis is perpendicular to plane of ring.	(2)	$\frac{MR^2}{12}$
(R)	 Uniform disk of initial mass M from which circular portion of radius R is then removed. M.I. of remaining mass about axis which is perpendicular to plane of plate	(3)	$\frac{13MR^2}{8}$

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

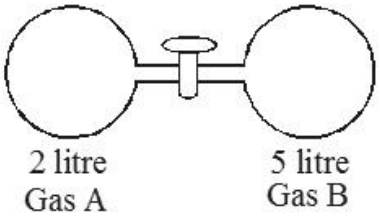
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. Two gases A & B are mixed under constant temperature condition. Initial volume of gas A is 2 litre & that of gas B is 5 litre. When stop cock is opened then which of the following statements is/are correct (initials moles of gas A = 2 & gas B = 5)-



(A) $\Delta S_{mix} = \ln\left[\left(\frac{7}{2}\right)^{2R} \times \left(\frac{7}{5}\right)^{5R}\right]$

(B) $\Delta S_{mix} = 0$, if gas 'A' & 'B' are same and equal volumes of them are mixed

(C) $\Delta S_{mix} = (n_1 + n_2)$ R ln2 if equal volumes of same gases are mixed

(D) ΔS_{mix} is maximum when equal moles of two gases are mixed. (Gas 'A' & 'B' are different)

2. Which of the following statement is/are correct?

(A) Sorel cement contains MgO & $MgCl_2$

(B) Silicates of calcium are important ingredients of portland cement

(C) Be & Mg dissolve in liquid ammonia to give deep blue black solution

(D) K_2CO_3 and Na_2CO_3 is used as fusion mixture in laboratory

Space for Rough Work

3. Which of the following statements are correct?

(A) $K_2Cr_2O_7$ is deliquescent but $Na_2Cr_2O_7$ is not a deliquescent

(B) In CrO_4^{2-} & $Cr_2O_7^{2-}$ geometry around Cr is same

(C) CrO_4^{2-} & $Cr_2O_7^{2-}$ produce orange & yellow colour respectively due to charge transfer

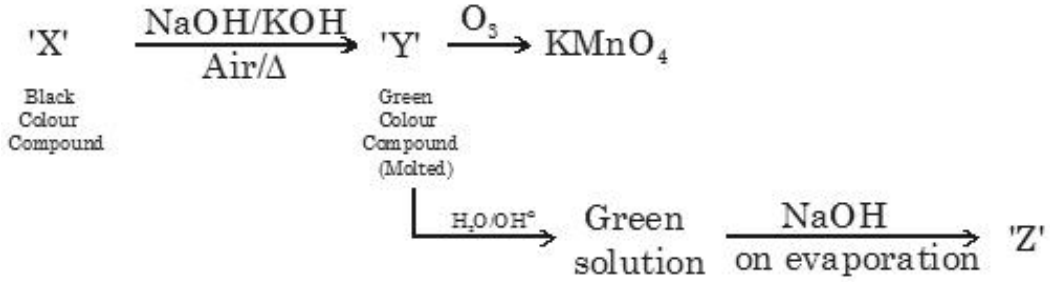
(D) In Deacon process $CuCl_2$ is used as a catalyst in making Cl_2 from HCl

4. Which of the following compound may be used to remove temporary hardness?

(A) $Ca(OH)_2$ (B) $Na_6P_6O_{18}$

(C) $Na_2Al_2Si_2O_8.xH_2O$ (D) $Na_2CO_3.10H_2O$

5. Consider the following reaction :-



Which of the following statement is/are correct?

(A) 'Z' is isomorphous with $Na_2SO_4.10H_2O$

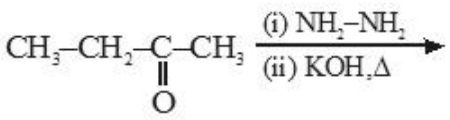
(B) In basic medium $MnO_4^{\ominus}$ produce 'X' in the form of brown ppt. when it is act as oxidising agent

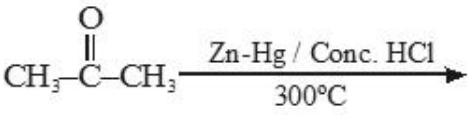
(C) 'Y' produce 'X' in the form of brown ppt. & $KMnO_4$ with water

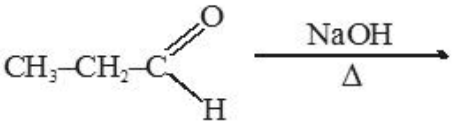
(D) 'Y' produce 'X' in the form of brown ppt. & $KMnO_4$ with water & CO_2

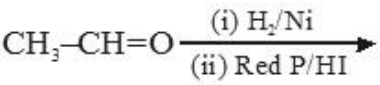
Space for Rough Work

6. Which one of the following reaction will produce hydrocarbon as final product?

(A) 

(B) 

(C) 

(D) 

7. Which Amino acid(s) is/are essential amino acid -

(A) Glycine

(B) Phenyl alanine

(C) Valine

(D) Threonine

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

Paragraph for Questions 8 - 10

Column-I	Column-II	Column-III
(A) $V_m = 50$ L	(P) $\frac{P}{RT} = \frac{1}{75}$	(I) $Z < 1$
(B) $V_m = 80$ L	(Q) $\frac{P}{RT} = \frac{1}{100}$	(II) $Z > 1$
(C) $V_m = 120$ L	(R) $\frac{P}{RT} = \frac{1}{40}$	(III) Attractive forces dominant
(D) $V_m = 20$ L	(S) $\frac{P}{RT} = \frac{1}{140}$	(IV) Molecular size dominant

1. If rate of diffusion of a real gas is 2/5 times of rate of diffusion of He and density of real gas = 0.5 gm/l then which is incorrect matching

(A) A, P, I (B) A, Q, I (C) A, R, I (D) A, P, III

Space for Rough Work

2. If molar mass of a real gas is 450 & density is 3.75 gm/litre then which is correct matching

(A) C, P, I (B) C, Q, I (C) C, S, IV (D) C, R, II

3. If molar mass of a real gas is 500 & 10 litre it has a weight of 62.5 gms then which INCORRECT matching

(A) B, P, II (B) B, R, IV (C) B, S, I (D) B, R, III

Paragraph for Questions 11 - 13

	Column-I (Carbohydrate)		Column-II (Monomer unit)		Column-III (Nnature of carbohydrate)
(P)	Cellulose	(i)	α -Glucose	(1)	Reducing sugar
(Q)	Sucrose	(ii)	β-Glucose	(2)	Non reducing
(R)	Starch	(iii)	Glucopyranose & Fructofuranose	(3)	Mono-saccharides
(S)	Lactose	(iv)	Glucose & β-Galactose	(4)	Poly-saccharides

4. Which of the following combination is correct for Cellulose ?

(A) ii , 2 , 4 (B) i , 2 , 4 (C) iv , 4 (D) ii , 2 , 3

5. Which of the following is not correctly match with their monomer ?

(A) P , ii (B) Q , iii (C) R , i (D) S , iii

Space for Rough Work

6. Which of the following is non reducing sugar ?

(A) α-Glucose (B) β-Glucose (C) Galactose (D) Starch

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

MATCHING LIST TYPE QUESTION (ANSWER AS THREE DIGIT INTEGER)

This section contains **FIVE (05)** questions.

The answer to each question is a **THREE DIGIT INTEGER** ranging from 111 to 999, both inclusive.

Each question having two matching lists. Choices for the correct combination of elements form List-I (P), (Q), (R) and List-II (1), (2), (3).

Given matching type questions in which List-I (P,Q,R) match with List-II (1,2,3) and represent your answer as four digit integer for example correct match for P ⟶ 2, Q ⟶ 3, R ⟶ 1.

Each matching in List-II may be/may not be correct & can be used more than one time for matching in **List-I**. For example your answer can be 223, 133, 143 etc...

For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

---

1. 1 mole ideal gas undergoes change in state from

2 atm, 5 litre, 300K) to (P_f , Vf & T_f)

(P) If w = –6.93 atm - litre (1) Process is irrreversible isothermal at P_{ext}= 1atm

(Q) If w = – 5 atm - litre (2) Process is isobaric at 2 atm pressure with V_f = 20 litre

(R) If w = – 30 atm - litre (3) Process is reversible isothermal with P_f = 1 atm

2.

Gases	A(g)	B(g)	C(g)	D(g)	E(g)	F(g)
$\Delta H^0_{fkJ/mole}$	40	60	30	20	80	100

(P) A(g) ⟶ B(g) + C(g) (1) Spontaneous at all temperature

(Q) C(g) + E(g) ⟶ D(g) (2) Spontaneous at low temperature

(R) F(g) ⟶ B(g) + C(g) (3) Spontaneous at high temperature

Space for Rough Work

3. Match given properties with elements

	List-I (Properties)		List-I (Elements)
P	Produce stable M_2O type oxide with limited supply of air	1	Li, Be, Al
Q	Chloride of elements produce red colour in oxidising flame	2	Li, Ca, Sr
R	Chlorides of such elements having significant covalent character or covalent nature in anhydrous form	3	Na, Li, Cs

4. Match the colour with given compounds

	List-I (Compounds)		List-II (Colour)
P	Mn_2O_7	1	Orange
Q	$(NH_4)_2Cr_2O_7$	2	Green
R	CrO_2Cl_2	3	Reddish Brown

5.

	List-I		List-II
P	Synthetic condensation copolymer	1	Buna-N
Q	Distinguish between aldehyde & ketones	2	Terylene
R	Addition copolymer	3	Tollen's reagent

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. Let $M=\begin{pmatrix}1 & 3\\ 0 & 2\end{pmatrix}$ and $I + C_1M + C_2M^2 = \begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}$, then answer the following ($C_1, C_2 \in R$ and I is an identity matrix of order '2')

(A) $C_1 = \frac{1}{2}$ (B) $C_1 = -\frac{3}{2}$ (C) $C_2 = \frac{1}{2}$ (D) $C_2 = -\frac{3}{2}$

2. Let $A=\begin{vmatrix} a & a+d & a+3d\\ a+d & a+2d & a\\ a+2d & a & a+d\end{vmatrix}$, then

(A) A = 0 (B) A depends on a

(C) A depends on d (D) A is independent of a and d

3. Area bounded by the curve $2|x| + 3|y| \le 6$ is n, then

(A) $n \ge 12$ (B) $n \le 10$ (C) $n > 12$ (D) $n > 10$

4. Let $f(x)=\begin{cases}x^{2\alpha+1}\ell nx & x>0\\ 0 & x=0\end{cases}$. If $f(x)$ satisfies rolle's theorem in [0,1], then α can be-

(A) $-\frac{1}{2}$ (B) $-\frac{1}{3}$ (C) $-\frac{1}{4}$ (D) –1

5. Let $A=\begin{pmatrix}2 & 2 & 1\\ 2 & 5 & 2\\ 1 & 2 & 2\end{pmatrix}$ and $B=\begin{pmatrix}-x & -y & z\\ 0 & y & 2z\\ x & -y & z\end{pmatrix}$ where $x,y,z \in R$. If $B^TAB = \begin{pmatrix}8 & 0 & 0\\ 0 & 27 & 0\\ 0 & 0 & 42\end{pmatrix}$, then xyz can be

(A) 4 (B) –4 (C) 6 (D) –6

Space for Rough Work

6. Consider two curves $y = e^x$, $y = e^{a-x}$, where a is positive number. Let S be the area of the part surrounded by C_1, C_2 and the y-axis, if $\lim\limits_{a\to 0} \frac{S}{a^2}$ is $\frac{1}{n}$, then

(A) n is the first even composite number

(B) number of positive divisors of n is 3

(C) n is a perfect square

(D) n is an odd number

7. The straight line which is both tangent and normal to the curve $x = 3t^2$, $y = 2t^3$ is

(A) $y + \sqrt{3}(x - 1) = 0$

(B) $y - \sqrt{3}(x - 1) = 0$

(C) $y + \sqrt{2}(x - 2) = 0$

(D) $y - \sqrt{2}(x - 2) = 0$

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

Paragraph for Questions 8 - 10

Column-I represents 2 curves.

Column-II Angle of intersection of between given curves.

Column-III Number of points of intersection of given curves.

Column-I		Column-II		Column-III	
(I)	$y = \sin x$ and $y = \cos x$ both in $(0,2\pi)$	(i)	$\tan^{-1}2\sqrt{2}$	(P)	2
(II)	$x^2 = 4y,\ y = \frac{8}{x^2+4}$	(ii)	$\frac{\pi}{2}$	(Q)	4
(III)	$\frac{x^2}{18} + \frac{y^2}{8} = 1,\ y = 0$	(iii)	$\tan^{-1}3$	(R)	0
(IV)	$xy = 1,\ x^2 - y^2 = 5$	(iv)	$\tan^{-1}5$	(S)	8

1. Which of the following options is the only **CORRECT** combination ?

(A) (I) (i) (P) (B) (II) (ii) (P) (C) (III) (iii) (P) (D) (IV) (iv) (P)

Space for Rough Work

2. Which of the following options is the only **CORRECT** combination ?

(A) (I) (ii) (P) (B) (II) (iv) (P) (C) (III) (ii) (P) (D) (IV) (iv) (Q)

3. Which of the following options is the only **CORRECT** combination ?

(A) (I) (ii) (P) (B) (I) (ii) (Q) (C) (I) (iii) (Q) (D) (IV) (ii) (P)

Paragraph for Questions 11 - 13

Column-I : y = $f(x)$ with respective domain.

Column-II : Interval in which $f(x)$ is increasing

Column-III : Interval in which $f(x)$ is decreasing

Column-I		Column-II		Column-III	
(I)	$y = \sin x + \cos x$ in $\left(0, \frac{\pi}{2}\right)$	(i)	$\left(0, \frac{\pi}{4}\right)$	(P)	$\left(0, \frac{\pi}{4}\right)$
(II)	$y = \sin 2x$ in $\left(0, \frac{\pi}{2}\right)$	(ii)	$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$	(Q)	$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
(III)	$y = \sin^5 x + \cos^5 x - 1$ in $\left(0, \frac{\pi}{2}\right)$	(iii)	$\left(0, \frac{\pi}{3}\right)$	(R)	$\left(0, \frac{1}{e}\right)$
(IV)	$y = x^x$ in $\left(0, \frac{\pi}{2}\right)$	(iv)	$\left(\frac{1}{e}, \frac{\pi}{2}\right)$	(S)	$\left(\frac{1}{e}, \frac{\pi}{2}\right)$

4. Which of the following options is the only **CORRECT** combination ?

(A) (I) (i) (S) (B) (II) (iii) (S) (C) (III) (ii) (P) (D) (I) (ii) (P)

5. Which of the following options is the only **CORRECT** combination ?

(A) (I) (iii) (P) (B) (II) (ii) (P) (C) (III) (iii) (P) (D) (IV) (iv) (R)

6. Which of the following options is the only **CORRECT** combination ?

(A) (I) (iv) (R) (B) (II) (i) (Q) (C) (III) (iii) (Q) (D) (II) (ii) (P)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

MATCHING LIST TYPE QUESTION (ANSWER AS THREE DIGIT INTEGER)

This section contains **FIVE (05)** questions.

The answer to each question is a **THREE DIGIT INTEGER** ranging from 111 to 999, both inclusive.

Each question having two matching lists. Choices for the correct combination of elements form List-I (P), (Q), (R) and List-II (1), (2), (3).

Given matching type questions in which List-I (P,Q,R) match with List-II (1,2,3) and represent your answer as four digit integer for example correct match for P $\longrightarrow$ 2, Q $\longrightarrow$ 3, R $\longrightarrow$ 1.

Each matching in List-II may be/may not be correct & can be used more than one time for matching in **List-I**. For example your answer can be 223, 133, 143 etc...

For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

---

1. Let $\lim\limits_{T\to\infty}\frac{1}{T}\int\limits_{0}^{T}(\sin x+\sin ax)^2dx = L$, then

List-I		List-II	
(P)	for a = 0 the value of L is	(1)	0
(Q)	for a = 1 the value of L is	(2)	$\frac{1}{2}$
(R)	for a = –1 the value of L is	(3)	2

Space for Rough Work

2.

List-I		List-II	
(P)	Value of definite integral $\int_{-1}^{1} \frac{dx}{\left(1+x^3+\sqrt{1+x^6}\right)}$	(1)	3
(Q)	Let $f(x) = \frac{x-1}{x^2}$; $x = k$ is point of inflection, then k is	(2)	1
(R)	$\int_{1}^{e} \frac{dx}{\ell n\left(x^x e^x\right)}$ equals	(3)	$\ell n2$

3.

List-I		List-II	
(P)	$\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma =$	(1)	2
(Q)	If $A = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}$ and a matrix is defined as $C = BAB^{-1}(B^{-1}A^TB)$, then detC is equal to	(2)	4
(R)	If $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ and $A^4 = -\lambda I$, then λ equals to	(3)	16

Space for Rough Work

4.

List-I		List-II	
(P)	If $I = \int_0^1 \ell n\left(\frac{1}{x} - 1\right)$, then I equals to	(1)	$\frac{1}{3}\ell n9$
(Q)	$\int_0^{10\pi} \left(\lim_{x \to y}\left(\frac{\sin x - \sin y}{x - y}\right)\right) dy$ is	(2)	0
(R)	$\lim_{n \to \infty}\left(\frac{1^2}{1^3 + n^3} + \frac{2^2}{2^3 + n^3} + \ldots + \frac{4}{9n}\right)$ equal to	(3)	1

5.

List-I		List-II	
(P)	A curve passing through (2,3) having the property that length of the radius vector of any of point P is equal to the length of tangent drawn at this point, curve can be	(1)	Straight line
(Q)	A curve passing through (1,1) having the property that any tangent intersects the y-axis at the point which is equidistant from the point of tangency and the origin curve can be	(2)	Circle
(R)	A curve passing through (1,0) for which length of normal is equal to radius vector, curve can be	(3)	Parabola

Space for Rough Work

Space for Rough Work

Topic :

PHYSICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

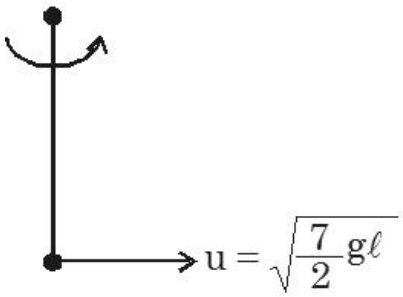
---

1. In certain electronic transition from quantum level n to ground state in atomic hydrogen in one or more steps no line belonging to Brackett series is observed. The wave numbers which may be observed in Balmer series is :-

(A) $\frac{8R}{9}, \frac{5R}{36}$ (B) $\frac{3R}{16}, \frac{8R}{9}$

(C) $\frac{5R}{36}, \frac{3R}{16}$ (D) $\frac{3R}{4}, \frac{3R}{16}$

2. A small ball is given a velocity $\sqrt{\frac{7}{2}g\ell}$ at the lowest point of verticle circle. Find the maximum height reached by the ball :-



(A) $\frac{4\ell}{5}$ (B) $\frac{5\ell}{4}$

(C) $\frac{3\ell}{2}$ (D) $\frac{27\ell}{16}$

3. In Bohr's theory for hydrogen-like atoms

(A) For the higher atomic number, radius of the nth orbit is higher.

(B) For the higher quantum number, velocity of electron in that orbit is greater.

(C) In some hypothetical condition if mass of electron would have been higher then electron velocity in that orbit would have been same.

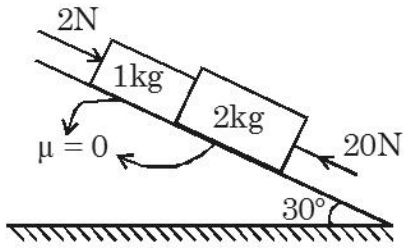
(D) For the higher quantum number, time period of an electron revolving around nucleus is smaller.

4. The K, L and M energy levels of platinum lie roughly at 78, 12 and 3 keV respectively. The ratio of wavelength of K_α line to that of K_β line in X-ray spectrum is :

(A) $\frac{22}{3}$ (B) $\frac{3}{22}$

(C) $\frac{22}{25}$ (D) $\frac{25}{22}$

5. The normal reaction force between the blocks and direction of acceleration are :-



(A) 8 N, up the incline

(B) 8N, down the incline

(C) 5N, up the incline

(D) 5N, down the incline

6. A normal stationary lithium atom releases an alpha particle with velocity $\left(2\hat{i}+3\hat{j}+7\hat{k}\right)$ km/s. What will be the velocity of the daughter element just after the release of the alpha particle?

(A) $(1/2)\left(2\hat{i}+3\hat{j}+7\hat{k}\right)$ km/s

(B) $(-4/3)\left(2\hat{i}+3\hat{j}+7\hat{k}\right)$ km/s

(C) $(-1/4)\left(2\hat{i}+3\hat{j}+7\hat{k}\right)$ km/s

(D) $(1/5)\left(2\hat{i}+3\hat{j}+7\hat{k}\right)$ km/s

7. The emitted $_{-1}\beta^0$ particle by a radioactive atom is an electron which is-

(A) Emitted from the orbit of the atom.

(B) Already present in the nucleus.

(C) Produced by the conversion of neutron into proton in the nucleus.

(D) Produced by the conversion of proton into neutron in the nucleus.

8. The radioactive sources A and B of half lives of 2hr and 4hr respectively, initially contain the same number of radioactive atoms. At the end of 2 hours, their rates of disintegration are in the ratio :-

(A) 4 : 1 (B) 2 : 1

(C) $\sqrt{2}$: 1 (D) 1 : 1

9. With the x-axis horizontal and the y-axis vertically upward as positive, the change in the horizontal component of velocity, Δv_x, and the change in the vertical component of velocity, Δv_y, of a projectile are related to the time since leaving the barrel, Δt, as

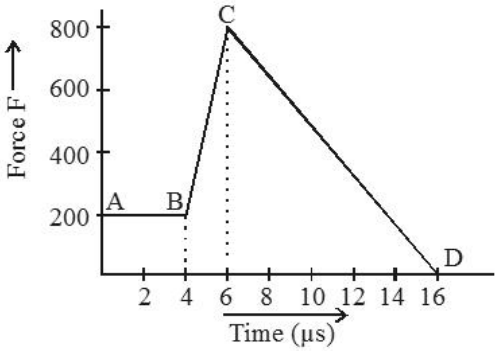
(A) $\Delta v_x = 0; \Delta v_y = 0$

(B) $\Delta v_x = g\Delta t; \Delta v_y = 0$

(C) $\Delta v_x = 0; \Delta v_y = g\Delta t$

(D) $\Delta v_x = 0; \Delta v_y = -g\Delta t$

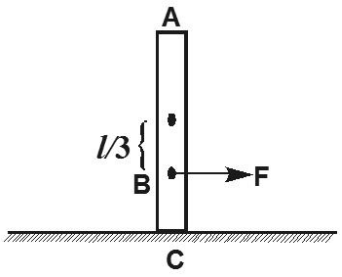
10. The net force (in newtons) acting on a body of mass 2kg (at rest) varies with time t (in microseconds) as shown in the figure. AB, BC and CD are straight line segments. Find velocity of the block at 16μs.



(A) 2.9×10^{-3} m/s (B) 2.9×10^{3} m/s

(C) 5×10^{-3} m/s (D) None of these

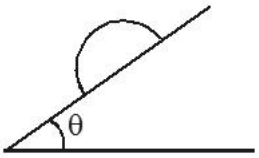
11. A uniform thin rod ABC of mass M and length ℓ is placed vertically on a rough horizontal surface. The coefficient of kinetic friction between the rod and the surface is μ. A force F (sufficient for sliding) is applied on the rod at point B at distance $\ell/3$ below centre of the rod as shown in figure. The initial acceleration of point A is



(A) $\mu g - \frac{F}{M}$ (B) $\frac{F}{M}$

(C) $4\mu g$ (D) $2\mu g - \frac{F}{M}$

12. A uniform semi circular ring is placed on rough inclined plane as shown. If friction is large for no sliding, then the minimum angle θ at which toppling occurs is :



(A) $\tan^{-1} 2$ (B) $\tan^{-1}\frac{\pi}{2}$

(C) 45° (D) $\tan^{-1}\frac{\pi}{4}$

13. As shown in figure, two blocks are connected with a light spring. When spring was at its natural length, velocities are given to them as shown in figure. Choose the wrong alternative.



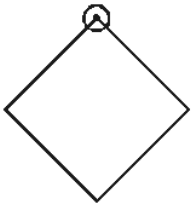
(A) Velocity of center of mass of the system is 3 m/s (towards right)

(B) When spring is maximum compressed velocity of 20 kg block is 3 m/s (towards right)

(C) When spring is maximum elongated velocity of 10 kg block is 3 m/s (towards left)

(D) At the time of maximum compression both blocks are moving in same direction.

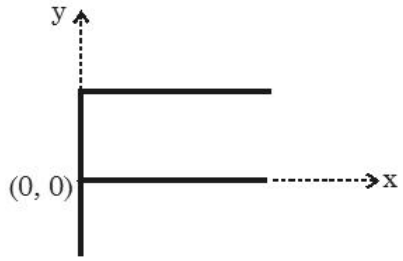
14. A square frame of mass m and side ℓ is suspended as shown. Find the time period of small oscillations in vertical plane :-



(A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{5\sqrt{2}}{6}\frac{\ell}{g}}$

(C) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$ (D) $2\pi\sqrt{\frac{5\sqrt{2}}{3}\frac{\ell}{g}}$

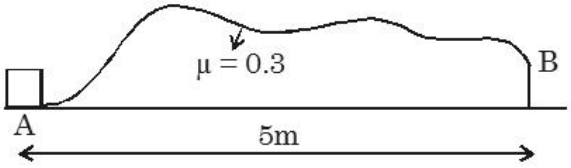
15. Letter F is formed from three identical rods of length ℓ. Find coordinates of its centre of mass.



(A) $\left(\frac{\ell}{5}, \frac{\ell}{5}\right)$ (B) $\left(\frac{\ell}{3}, \frac{\ell}{6}\right)$

(C) $\left(\frac{\ell}{6}, \frac{\ell}{3}\right)$ (D) $\left(\frac{\ell}{2}, \frac{\ell}{5}\right)$

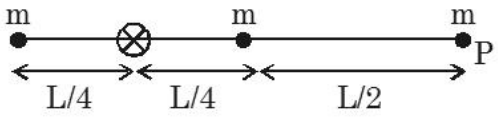
16. Find the work done by friction in slowly moving a block of mass 2kg from A to B on a small hill as shown in figure :-



(A) –30 J (B) 30 J

(C) –20 J (D) –15 J

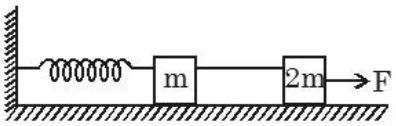
17. A uniform rod of mass 5m is hinged as shown. Three particles each of mass m are kept as shown. Find acceleration of particle P at t = 0 just after release :-



(A) $\frac{18g}{17}$ (B) $\frac{g}{4}$

(C) $\frac{g}{16}$ (D) $\frac{39}{16}g$

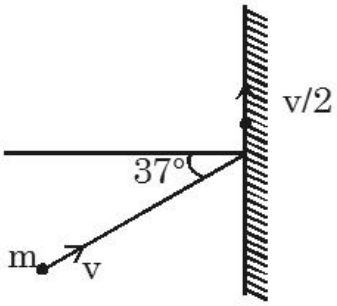
18. The system is in equilibrium with applied force F as shown. At t = 0 force F is removed. Initial acceleration of 2m block will be :-



(A) $\frac{F}{m}$ (B) $\frac{F}{3m}$

(C) $\frac{F}{2m}$ (D) $\frac{2F}{m}$

19. A ball collides with a wall and moves parallel to the wall as shown. The impulse acting on the ball during the collision is :-



(A) $mv\left(-\frac{4}{5}\hat{i} - \frac{1}{10}\hat{j}\right)$ (B) $mv\left(-\frac{4}{5}\hat{i} - \frac{2}{5}\hat{j}\right)$

(C) $mv\left(-\frac{3}{5}\hat{i} - \frac{1}{10}\hat{j}\right)$ (D) $mv\left(-\frac{3}{5}\hat{i} - \frac{2}{5}\hat{j}\right)$

20. A sample of H atoms in certain excited state emits 10 different wavelengths in de-excitation to ground state. The maximum wavelength out of these wavelengths is :-

(A) 4.05×10^{-6} m (B) 4.05×10^{-4} m

(C) 4.5×10^{-6} m (D) 4.5×10^{-4} m

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

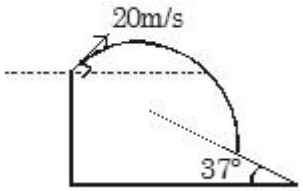
Zero Marks : 0 In all other cases.

---

1. A silver sphere (work function 4.6 eV) is suspended in a vacuum chamber by an insulating thread. Ultraviolet light of wavelength 0.2 μ m strikes on the sphere. The maximum electric potential (in Volts) of the sphere will be (hc = 12400 eVÅ) :-

2. Rain appears to be falling at 53° with the vertical to a man standing on the ground. When he starts running at 10 km/hr, rain again appears to be falling at 53° with the vertical. Find the speed (in km/hr) of rain wrt ground :-

3. A projectile is thrown perpendicular to a long inclined plane as shown. The range (in m) of projectile on the incline is :-



4. The wavelength of K_α X-ray of an element having atomic number Z = 13 is λ. The wavelength of K_α X-ray of another element of atomic number Z' is 4λ. Then Z' is

5. A sample of radioactive nuclei of a certain element can decay only by γ-emission and β-emission. If the half-life for γ-emission is 24 minutes and that for β-emission is 36 minutes, the half-life (in minute) for the sample is

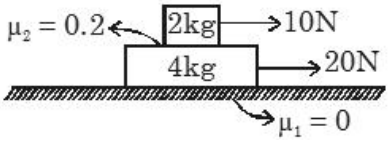
6. A block of metal weighing 2 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 kg/s and at a speed of 5 m/s. The initial acceleration (in m/s^2) of the block will be (Assume water doesn't rebound) :



7. A ball is thrown with 50 m/s at 30° with the horizontal from a building of height 500 m. The time (in sec) at which velocity vector of the ball becomes perpendicular to the initial velocity is (g = 10 m/s^2) :-

8. A particle starts from rest and performs SHM of amplitude A. Find the ratio of time taken by it from mean to $\frac{A\sqrt{3}}{2}$ to time taken by it from A to $\frac{A}{2}$:-

9. If there will be no relative motion between the two blocks, the friction force (in N) between the blocks will be :-



10. A ball is thrown with $40\sqrt{2}$ m/s at 45° with horizontal. Find the radius of curvature (in m) of its path at t = 7 sec :-

Topic :

CHEMISTRY

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

1. Which of the following is INCORRECT statement ?

(A) In adiabatic expansion of an ideal gas, temperature of system always decrease

(B) In isothermal expansion of an ideal gas, temperature of a gas always remains constant

(C) In a reversible adiabatic compression of an ideal gas, change in entropy of system is zero

(D) In an irreversible adiabatic compression of an ideal gas, change in entropy of system is positive

2. Which of the following arrangement does not represent correct increasing order of the property stated against it?

(A) Sc Ti V Cr : Highest possible oxidation number

(B) Ti V Cr Mn : Melting point

(C) Cu Ni Co Fe : Paramagnetic behaviour

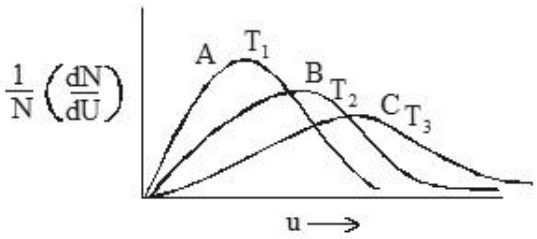
(D) Mn Fe Co : E° (M^{2+}/M) standard reduction potential

3. Bone deformities in children is due to deficiency of which vitamins :

(A) Ascorbic acid (B) Thiamine

(C) Riboflavin (D) Vitamin-D

4.



Which of the following statement is correct.

(i) If A & B are same gases then $T_2 > T_1$

(ii) If $T_1 = T_2$ then $A_{(g)}$ may be $SO_{2(g)}$ & $B_{(g)}$ may be $CH_{4(g)}$

(iii) If B & C are same gases then $T_3 > T_2$

(iv) If $T_1 = T_2 = T_3$ then $A_{(g)}$ may be CH_4, $B_{(g)}$ may be SO_2 & $C_{(g)}$ may be $SO_{3(g)}$

(A) i, ii & iii only

(B) i, ii, & iv only

(C) All are correct

(D) ii & iii only

5. Which of the following is electron precise molecular hydride ?

(A) B_2H_6 (B) SiH_4

(C) NH_3 (D) BeH_2

6. In which process hardness of water is removed by the formation of water soluble complex?

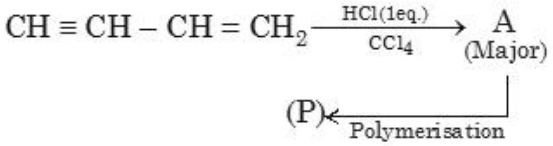
(A) Calgon's process

(B) Zeolite method

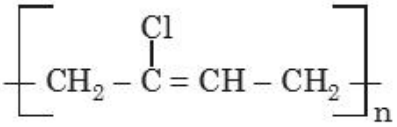
(C) Clark's method

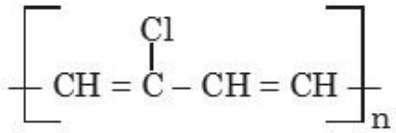
(D) Washing soda method

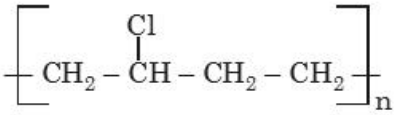
7.

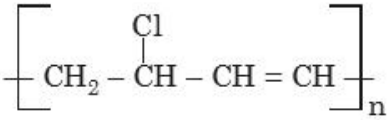


Product P is :

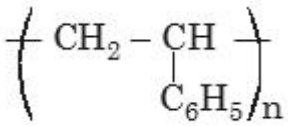
(A) 

(B) 

(C) 

(D) 

8. Identify the monomer in the given polymeric structures :



(A) Propene

(B) Styrene

(C) Vinylchloride

(D) Ethene

9. For which of the following reaction entropy, change in the system would be positive (in forward direction)

(A) $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

(B) $3Mg_{(s)} + 2Fe^{3+}_{(aq)} \rightarrow 3Mg^{2+}_{(aq)} + 2Fe_{(s)}$

(C) $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$

(D) $C_{(graphite)} \rightarrow C_{(diamond)}$

10. Misch metal consist of :-

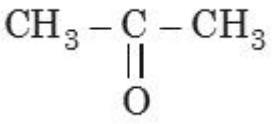
(A) ~95% Lanthanoides and ~5% Fe

(B) ~5% Lanthanoides and 95% Fe

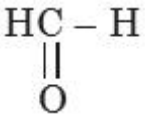
(C) ~95% Actinides and 5% Al

(D) ~95% Lanthanoides and ~5% Al

11. Which of the following aldehyde gives positive Iodoform test?

(A) 

(B) $CH_3 - CH = O$

(C) 

(D) $Me_3C - CH = O$

12. Which of the following is non reducing sugar ?

(A) Maltose (B) Sucrose

(C) Fructose (D) Lactose

13. For which of the following process $\Delta G = 0$

(A) $A_{(l)} \rightarrow A_{(g)}$ (1 atm, 90°C), B.P. of $A_{(l)}$ at 1 atm = 90°C

(B) $H_2O_{(l)} \rightarrow H_2O_{(g)}$ (1 atm, 400°C)

(C) $H_2O_{(l)} \rightarrow H_2O_{(s)}$ (1.0 atm, –10°C)

(D) $A_{(l)} \rightarrow A_{(g)}$ (2 atm, 90°C), B.P. of $A_{(l)}$ at 1 atm = 90°C

14. Select the **CORRECT** statement for heavy water.

(A) Heavy water has lower boiling point than H_2O

(B) Heavy water is more effective solvent than ordinary water

(C) Heavy water is more associated than ordinary water

(D) Ice cube of heavy water floats on H_2O(liq.)

15. Incorrect statement regarding solution obtained by dissolving alkali metals in liquid ammonia.

(A) Paramagnetic nature

(B) Show oxidising nature

(C) Deep blue due to presence of ammoniated electrons

(D) Good conductor

16. Which of the following reaction is not correctly matched with their major product?

(A) $CH_3CHCHCH_2CN \xrightarrow[(ii)\,H_2O]{(i)\,AlH(i-Bu)_2} CH_3CHCHCH_2CHO$

(B) $CH_3CH_2CN \xrightarrow[(2)\,H_3O^+]{(i)\,SnCl_2\,/\,HCl} CH_3CH_2CHO$

(C) $CH_3CH_2COCl + (CH_3)_2Cd \rightarrow CH_3COCH_3 + CdCl_2$

(D) $C_2H_5COCH_3 \xrightarrow{NaOCl} C_2H_5COO^- Na^+ + CHCl_3$

17. Select the correct statement :-

(i) Lithium nitrate when heated gives N_2 gas

(ii) LiF and Li_2O are comparatively much less soluble in water than the corresponding compound of alkali metals

(iii) LiCl is deliquescent and crystallises as a hydrate, $LiCl.2H_2O$

(iv) In KO_2, peroxide ion is present which is diamagnetic

(A) (i), (ii) (B) (ii),(iii)

(C) (i),(iv) (D) (iii),(iv)

18. Select the correct statement :-

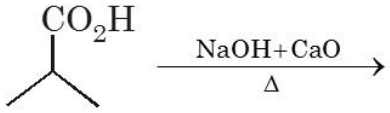
(A) MnO_2 is used in dry battery cells

(B) Lanthanoids react with H_2O to formed $Ln(OH)_4$ and H_2

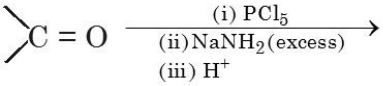
(C) Lanthanoids react with acids & do not produce H_2

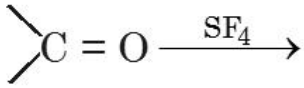
(D) In actinoids series common oxidation state is +4

19. In which reaction hydrocarbon is not formed as major product :

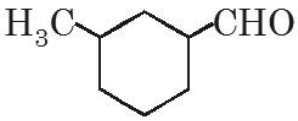
(A) 

(B) $Me_3C-Cl \xrightarrow[\text{dry ether}]{\text{Na in}}$

(C) 

(D) 

20. Correct IUPAC name of given compound is



(A) 3-methylcyclohexanecarbaldehyde

(B) 3-methylcyclohexanal

(C) 5-Methylcyclohexane carbaldehyde

(D) 3-Methylcyclohexanol

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

---

1. For the real gases reaction

$2A_{(g)} + B_{(g)} \rightarrow 4C_{(g)}$; $\Delta H = ?$

In a 10 litre vessel at 500 K the initial pressure is 50 bar. After the complete reaction, volume of gases was found to be 15 litre at a pressure of 80 bar. Find change in enthalpy (in kJ) if change in internal energy is +50 kJ {1 bar-litre = 100 Joule}

2. In acidic medium $KMnO_4$ is decolourised by which of the following specie(s). Give the total sum of the atoms present in them :-

(a) $H_2C_2O_4$ (b) HNO_3

(c) $Na_2S_2O_3$ (d) HNO_2

3. Which of the following is essential amino acid? Give total number of hydrogen atoms present in it.

(a) Proline (b) Glutamine

(d) Serine (c) Valine

4. 50 gm $H_2O(A)$ is mixed with 70 gm of $H_2O(B)$ in a rigid adiabatic vessel. The initial temperature of A & B was 46° & 70°C. Find ΔH_A [change in enthalpy of water $H_2O(A)$] (in calories) specific heat of water = 1 cal/gm/°c

5. $HCl + BOH \rightarrow BCl + H_2O \quad \Delta H = -57$ kJ

$CH_3COOH + BOH \rightarrow CH_3COOB + H_2O \quad \Delta H = -50$ kJ

$HCl + DOH \rightarrow DCl + H_2O \quad \Delta H = -45$ kJ

then ΔH (in kJ) for $CH_3COOH + DOH \rightarrow CH_3COOD + H_2O$, is equal to :-

6. In one of the following reaction H_2O_2 shows reducing nature. Find the n-factor of the species getting reduced ?

(1) $CrO_4^{2-} + 2H_2O_2 \xrightarrow{H^+} CrO_5 + 3H_2O$

(2) $MnO_2 + H_2O_2 + H_2SO_4 \longrightarrow MnSO_4 + 2H_2O + O_2$

(3) $H_2S + H_2O_2 \longrightarrow S\downarrow + 2H_2O$

(4) $Mn^{+2} + H_2O_2 \xrightarrow{OH^-} MnO_2 + 2H_2O$

7. The molecular speed in a sample of 100 molecules are distributed as

Number of molecules

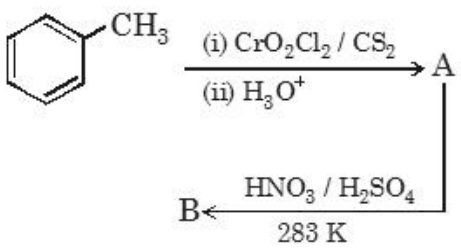
10 20 40 15 10 5

Speed m/sec.

60 80 100 120 140 160

Give the sum of V_{mp} & $V_{avg.}$ in m/sec

8.



Give the total number of atoms present in final product 'B'.

9. The critical point of CO_2 is 304 K and 73 atm. Under which condition is carbon dioxide liquid.

I) 305 K, 70 atm II) 303 K, 75 atm

III) 303 K, 73 atm IV) 305 K, 80 atm

Give the ratio

$$\frac{\text{sum of temperature (in K) of all these conditions}}{\text{sum of pressure (in atm) of all these conditions}}$$

10. At what temperature (in K) is the following reaction at equilibrium, when all substances are at standard pressure? Assuming that entropies and enthalpies of reaction do not vary with temperature.

$PCl_3 + Cl_{2(g)} \rightleftharpoons PCl_{5(g)}$

	ΔH_f^0 kJ/mole	S°J/mole/kelwing
$PCl_{3(g)}$	–290	310
$Cl_{2(g)}$	0	220
$PCl_{5(g)}$	–375	360

Topic :

MATHEMATICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

1. The value of $\int_0^{10\pi} [\sin x + \cos x]\, dx$ is (where [.] denotes greatest integer function)

(A) -5π (B) -10π

(C) $-\frac{5\pi}{2}$ (D) $-\frac{15\pi}{2}$

2. Let $f : R \to R$ be a continuous function. If $f(x) + f(x+1) + f(x+2) + f(x+3) = 2021$, $\forall\, x \in R$ and $\int_0^4 f(x)\,dx = A$, then the value of $\int_0^{100} f(x)\,dx$ is equal to -

(A) A (B) 20A

(C) 25A (D) 50A

3. If $f(x) = 2(g(x))^3 - 3(g(x))^2 + 3g(x) - 2x + \cos x$, where g(x) is a differentiable function then which of the following is always true-

(A) $f(x)$ is increasing if g(x) is decreasing

(B) $f(x)$ is decreasing if g(x) is increasing

(C) $f(x)$ is increasing if g(x) is increasing

(D) $f(x)$ is decreasing if g(x) is decreasing

4. Angle between the tangents drawn at origin to the two curves $x^5 + 4x^3y^2 + 3x + 4y = 0$ and $x^3 - 27y^3 + 5(4x - 3y)^2 = 0$ is

(A) $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3$

(B) $\cot^{-1}1 + \cot^{-1}2 + \cot^{-1}3$

(C) is the angle between the lines $11y = 2x$ and $x + 2y = 0$

(D) $2\cot^{-1}1 + \cot^{-1}2$

5. The point on the curve $y^3 = 4x^2 + 12y$ at which the tangent is horizontal is

(A) (2,2) (B) (–2,–2)

(C) (0,0) (D) (1,1)

6. Equation of a straight line with gradient 1 which touches curve $y = \frac{x}{1+|x|}$ is

(A) $y = x$ (B) $y = x + 1$

(C) $y = x - 1$ (D) $y = x + 2$

7. If A is a symmetric matrix ($|A| = -1$) and $AA^T = I$, then identify the incorrect option-

(A) $A^T = A^{-1}$ (B) $A^T + adjA = 0$

(C) $A^2 = A^4$ (D) $A^2 = A^3$

8. The function $f(x) = \int_{-2}^{x} (e^{-t} - 1)(t^2 - 1)^5 (t+2)^9 (t-3)^7$ has a local minimum at 'x' equal to

(A) –2 (B) 0 (C) 3 (D) –1

9. If $f(x)$ is an even function such that $f(10 - x) = f(x)$ and $\int_0^5 f(x)dx = 7$. Then the value of $\int_0^{10} f(x)dx$ is

(A) 10 (B) 12

(C) 14 (D) 21

10. If $f(x) = \int (2\sin(2\ell nx) - \cos(2\ell nx))\,dx$ and $f\left(e^{\frac{\pi}{2}}\right) = 0$. Then the value of $f(e^{\pi/4})$ is

(A) $-e^{\pi/2}$

(B) $e^{\pi/4}\left(1 - 2e^{\pi/4}\right)$

(C) $e^{\pi/4}\left(2e^{\pi/4} - 1\right)$

(D) $e^{\pi/4}\left(2e^{\pi/2} - 1\right)$

11. If length of subnormal to the curve $x\left(\frac{y}{k}\right)^n = k$ is always constant then 'n' is equal to

(A) 2 (B) 1 (C) –1 (D) –2

12. Let A and B are orthogonal matrices of order 3 × 3 such that $X = AB^T$ and $Y = BA^T$, then the value of tr(XY) is equal to -

(A) 1 (B) 2 (C) 3 (D) 4

13. Number of point(s) of inflection of $f(x) = x^4 + 6x^3 + 35x^2 + 6x - 3$ is -

(A) 0 (B) 1 (C) 2 (D) 4

14. Solution of differential equation $\left(\sqrt{1 - x^2y^2} - y\right)dx = xdy$ cannot be

(A) $xy + \cos x = 0$ (B) $xy = \sin x$

(C) $xy + \sin x = 0$ (D) $xy = \tan x$

15. The curve which satisfy differential equation $(x^2y - y)dx + (xy^2 - x)dy = 0$ is-

(A) $\pi x^2y^2 = e^{x^2+y^2}$ (B) $\pi x^2y^2 = e^{x+y}$

(C) $exy = e^{x+y}$ (D) $exy = e^{x^2+y^2}$

16. A function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} + x^2y + 2x = 0$, $f(1) = 1$, then the value of $f'(1)$ is-

(A) 1 (B) –1 (C) 3 (D) –3

17. Equation of tangent drawn to the curve $f(x) = x^3\frac{(2^x - 1)}{(2^x + 1)} + \sin^2 x\cos^6 x - 1$ at its point of intersection with y-axis is-

(A) $y + 1 = 0$ (B) $x + 3y = 10$

(C) $x + y = 0$ (D) $x = -1$

18. Consider a system of equations $x - y + z = -1$, $\lambda x + 3y + 2z = 3$ and $3x + \lambda y + z = 2$, if system have a unique solution then λ **cannot** be-

(A) 3 (B) –3 (C) –4 (D) 2

19. If $A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$ also, $|adj\lambda A| = \lambda^a . a^\lambda$ (where $a, \lambda \in N$). Then the value of '$\alpha + \lambda$', is-

(A) 6 (B) 8 (C) 10 (D) 12

20. If $A = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$, then the value of $\det(A^{2019} - 3A^{2018})$ is equal to-

(A) –7 (B) 7 (C) 5 (D) –5

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

---

1. If $\int e^x \left(\frac{\tan x}{1+\tan x} + \frac{1}{1+\sin 2x} - \frac{1}{2} \right) dx = e^x f(x) + c$, where c is integration constant. Then $\left[5f\left(\frac{5\pi}{8} \right) \right]$ is (where [.] denotes greatest integer function)

2. If $f(x) = \begin{vmatrix} 3x^2 - 6x + 5 & 6x^2 + 6x - 3 & 3x^2 + 3x - 2 \\ 2x - 2 & 4x + 2 & 2x + 1 \\ 1 & 2 & 1 \end{vmatrix}$, then the value of $\frac{f(1) + f(3)}{f(2)}$

3. Let $f(x)$ be a thrice differentiable function such that it intersects the curve $g(x) = 2x^3 + 7x^2 - 5x + 1$ at exactly 7 distinct points, then minimum number of roots of $f'''(x) = 12$ is

4. The area bounded by $y^2 = 4x$ and $x^2 = 4y$ is

5. The area between the curve $y = 2x^4 - x^2$, the x-axis and the ordinates of two minima of the curve is of the form $\frac{a}{b}$ (where a and b are co-prime). Then the value of $(b - 10a)$ is-

6. If $f(x) = 2000x^3 - 3000x^2 - 36000x + 2021$ has local maximum at $x = p$ and local minimum at $x = q$, then the value of $\left|\frac{q}{p}\right|$ is

7. Area (in square units) bounded by the curves $f(x) = (\cos x + \sec x)^2$ and $g(x) = (\sin x + \operatorname{cosec} x)^2$ between $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$ is of the form $\frac{19\sqrt{b} - c}{d}$, s.t. H.C.F of b,c and d is 3, then the value of $\frac{b + c}{d}$ is-

8. The area bounded by pair of straight lines $x^2 + y^2 - 2xy - 5x + 5y + 6 = 0$ and the co-ordinate axes is-

9. Minimum distance between the curves $C_1 : x^2 + y^2 = 8$ and $C_2 : xy = 9$ is-

10. If $f(x) = x^3 + 3ax^2 + (11a - 6)x - 2$ is increasing function for all $x \in R$, then number of possible integral value(s) of 'a' is/are-

Space for Rough Work

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
27-06-2021

JEE(Main + Advance) : IIT-JEE(M+A) XII & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 27-06-2021

ANSWER KEY PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,C	C,D	A,C	A,D	B,D	A,B	A,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	C	D	C	B	A	
SEC-3	Q.	1	2	3	4	5		
	A.	112	213	321	312	213		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,D	A,B,D	B,D	A,B,C,D	A,B,C,D	A,B,D	B,C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	D	D	A	D	D	
SEC-3	Q.	1	2	3	4	5		
	A.	312	321	321	213	231		

MATHEMATICS

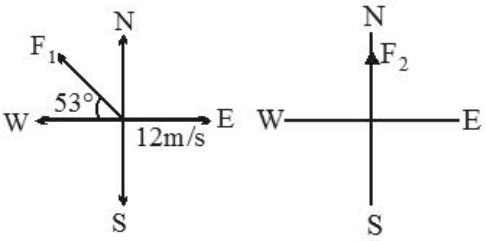
SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	B,C	A,D	B,C	C,D	A,B,C	C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	A	C	D	C	D	B	
SEC-3	Q.	1	2	3	4	5		
	A.	231	213	132	221	122		

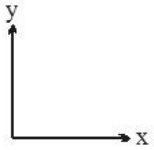
HINT – SHEET

PHYSICS

SEC-1

1. Ans (A,C)





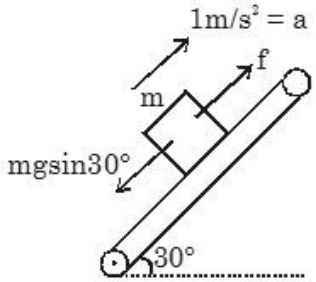
Vel. of air = $v\hat{j}$

$\vec{v}_{AB} = v\hat{j} - 12\hat{i}$

$\tan 53° = \frac{v}{12} = \frac{4}{3}$ $\quad \therefore v = 16$ m/s

$v_{AB} = \sqrt{(16)^2 + (12)^2} = 20$ m/s

2. Ans (C,D)



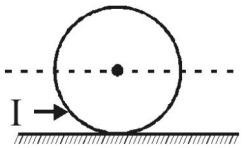
$N = mg\cos 30°$

$f_L = \mu N$

$f - mg\sin 30° = ma$

$f_L \geq f$

4. Ans (A,D)



$I = mv_0$

$I \times \frac{2R}{3} = \frac{2}{5}mR^2\omega$

$\omega = \frac{5I}{3mR}$

$= \vec{v} = \left(\frac{I}{m} + \frac{5I}{3m}\right)\hat{i}$

Velocity of point of contact is along positive x-axis so friction will be along negative x-axis.

6. Ans (A,B)

$\frac{n\lambda}{2} = L$

Particles energy in nth state

$E = \frac{P^2}{2m} = \frac{h^2}{2m\lambda^2} = \frac{n^2}{8m\ell^2}$

7. Ans (A,D)

$200 = 238 \times E - 238 \times 7.6$

$\frac{200}{238} + 7.6 = E$

PHYSICS

SEC-2

1. Ans (B)

Time after slipping ceases

$t = \frac{2v_0 m}{\mu mg\left(1 + \frac{mR^2}{I}\right)}$

$I_{Ring} > I_{Sphere} > I_{Disk} > I_{Solid\ sphere}$

$KE_i = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv_0^2 + \frac{1}{2}\frac{Iv_0^2}{R^2}$

$\Rightarrow KE_i = \frac{1}{2}v_0^2\left[m + \frac{I}{R^2}\right]$

$\Rightarrow KE_f = \frac{1}{2}v^2\left[m + \frac{I}{R^2}\right]$

from angular momentum conservation (LAMC)

$v = \frac{(mR^2 - I)v_0}{(mR^2 + I)}$

2. Ans (C)

Time after slipping ceases

$t = \frac{2v_0 m}{\mu mg\left(1 + \frac{mR^2}{I}\right)}$

$I_{Ring} > I_{Sphere} > I_{Disk} > I_{Solid\ sphere}$

$KE_i = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv_0^2 + \frac{1}{2}\frac{Iv_0^2}{R^2}$

$\Rightarrow KE_i = \frac{1}{2}v_0^2\left[m + \frac{I}{R^2}\right]$

$\Rightarrow KE_f = \frac{1}{2}v^2\left[m + \frac{I}{R^2}\right]$

from angular momentum conservation (LAMC)

$v = \frac{(mR^2 - I)v_0}{(mR^2 + I)}$

3. Ans (D)

Time after slipping ceases

$$t = \frac{2v_0 m}{\mu mg\left(1 + \frac{mR^2}{I}\right)}$$

$I_{Ring} > I_{Sphere} > I_{Disk} > I_{Solid\ sphere}$

$$KE_i = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv_0^2 + \frac{1}{2}\frac{Iv_0^2}{R^2}$$

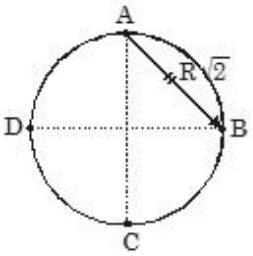
$$\Rightarrow KE_i = \frac{1}{2}v_0^2\left[m + \frac{I}{R^2}\right]$$

$$\Rightarrow KE_f = \frac{1}{2}v^2\left[m + \frac{I}{R^2}\right]$$

from angular momentum conservation (LAMC)

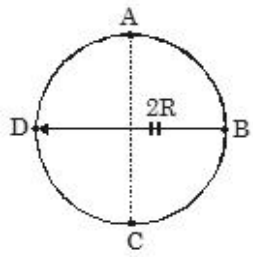
$$v = \frac{\left(mR^2 - I\right)v_0}{\left(mR^2 + I\right)}$$

4. Ans (C)



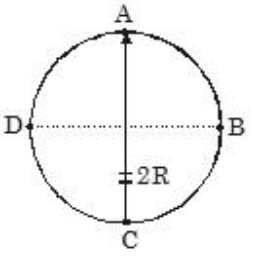
Δt = 1 sec

5. Ans (B)



Δt = 2 sec

6. Ans (A)



Δt = 2 sec

PHYSICS

SEC-3

3. Ans (321)

(A) Stopping potential in electron volts

$= hv - \phi = 12 - 4 = 8.$

(B) $\left(\frac{Z_1 - 1}{Z_2 - 1}\right)^2 = \frac{\lambda_2}{\lambda_1} = \left(\frac{85}{81}\right)^2$. Therefore

$Z_1 = 86$ and $Z_2 = 82$

(C) Half life time of radioactive material is 4 min. For 80 gm to reduce to 20gm, two half life times are required.

(D) The binding energy per nucleon for helium in MeV is approximately $\frac{0.0302 \times 930}{4} \approx 7$

4. Ans (312)

In x-direction	In y-direction
$u_x = +8$	$u_y = +8$ m/s
$a_x = -2$	$a_y = -2$m/s^2
$v_x = (8 - 2t)\hat{i}$	$v_y = (8 - 2t)\hat{j}$

(P) $v_x = 0$, at $t = 4$ $v_y = 0$, at $t = 4$ sec

(Q) $x = \frac{u_x^2}{2a_x} = 16$ m at $t = 4$ sec

(R) at $t = 0$ & $t = 8$ sec

$|\vec{v}_{rel}| = 8\sqrt{2}$ m/s

CHEMISTRY

SEC-1

2. Ans (A,B,D)

(i) Important ingredients present in portland cement are dicalcium silicate (Ca_2SiO_4)26%, tricalcium (Ca_3SiO_5)51% and tricalcium aluminate ($Ca_3Al_2O_6$)11%

(ii) Be & Mg does not produce blue colour in liquid ammonia..

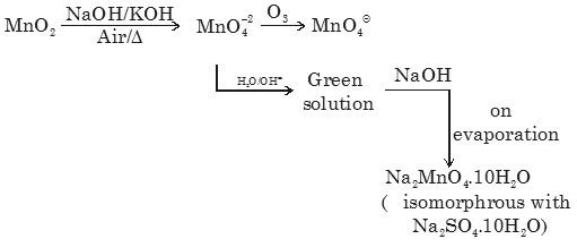
3. Ans (B,D)

(i) $Na_2Cr_2O_7$ is deliquescent & $K_2Cr_2O_7$ is not a deliquescent

(ii) In CrO_4^{2-} & $Cr_2O_7^{2-}$ geometry around Cr is tetrahedral

(iii) CrO_4^{2-} & $Cr_2O_7^{2-}$ are yellow & orange respectively due to charge transfer

5. Ans (A,B,C,D)



CHEMISTRY

SEC-3

1. Ans (312)

w for (1) = – 1 (10 – 5) = – 5 atm litre

w for (1) = – 2 (15) = – 30 atm litre

w for (1) = – 1 (1 × 10 × ln2 = – 10 × ln2 = – 6.93

2. Ans (321)

(A) $\Delta S > 0$; $\Delta H_{reaction} = +50$ kJ > 0

$\Delta G = 50 - T \times \Delta S$

So T should be high for $\Delta G < 0$

(B) $\Delta S < 0$; $\Delta H_{reaction} = -90$ kJ < 0

$\Delta G = -90 - T \times \Delta S$

So T should be low for $\Delta G < 0$

B – 2

(C) $\Delta S > 0$; $\Delta H = -10 < 0$

$\Delta G = -10 - T \times \Delta S$

$\Delta G < 0$ at all temperature

C – 1

MATHEMATICS

SEC-1

1. Ans (B,C)

$$M = \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$$

$$M^2 = \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 9 \\ 0 & 4 \end{pmatrix}$$

$$I + C_1M + C_2M^2 = \begin{pmatrix} 1+C_1+C_2 & 3C_1+9C_2 \\ 0 & 1+2C_1+4C_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

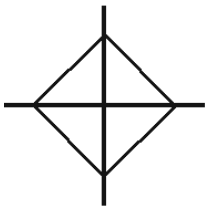
$C_1 + C_2 = -1$ $\quad 3C_1 + 9C_2 = 0$

$C_1 = -\frac{3}{2}$ $\quad C_2 = \frac{1}{2}$

2. Ans (B,C)

expand and comment

3. Ans (A,D)



$$A = \frac{1}{2} \times 6 \times 4 = 12$$

4. Ans (B,C)

$$\lim_{x \to 0^+} x^{2\alpha+1} \ell nx = 0$$

$2\alpha + 1 > 0$

$\alpha > -\frac{1}{2}$

5. **Ans (C,D)**

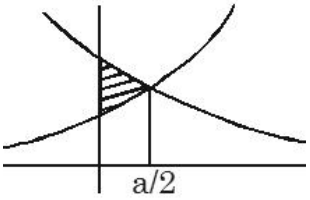
$$AB=\begin{bmatrix}-x & -y & 7z\\ 0 & y & 14z\\ x & -y & 7z\end{bmatrix}$$

$$B^TAB=\begin{pmatrix}2x^2 & 0 & 0\\ 0 & 3y^2 & 0\\ 0 & 0 & 42z^2\end{pmatrix}=\begin{pmatrix}8 & 0 & 0\\ 0 & 27 & 0\\ 0 & 0 & 42\end{pmatrix}$$

$2x^2 = 8, \quad 3y^2 = 27, \quad 42z^2 = 42$

$x = \pm 2, \quad y = \pm 3, \quad z = \pm 1$

6. **Ans (A,B,C)**

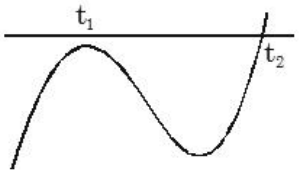


$$A=\int_0^{a/2}\left(e^{a-x}-e^x\right)dx=\left(e^{a/2}-1\right)^2$$

$$\lim_{a\to 0}\frac{e^{a/2}-1}{4\frac{a^2}{4}}=\frac{1}{4}$$

$n = 4$

7. **Ans (C,D)**



$$\frac{dy}{dx}\Big)_{t_1}=-\frac{1}{dx/dx}\Big)_{t_2}=\frac{2t_2^3-2t_1^3}{3\left(t_2^2-t_1^2\right)}$$

$$\frac{6t_1^2}{6t_1}=-\frac{6t_2}{6t_2^2}=\frac{2}{3}\frac{\left(t_2^2+t_1^2+t_1t_2\right)}{t_2+t_1}$$

$$t_1=-\frac{1}{t_2}=\frac{2}{3}\frac{t_2^2+t_1^2+t_1t_2}{t_1+t_2}$$

$t_1 = \pm\sqrt{2}$

Straight lines are

$y=-\sqrt{2}(x-2)$

$y=\sqrt{2}(x-2)$

MATHEMATICS

SEC-2

1. **Ans (A)**

(I) $y = \sin x \qquad m_1 = y' = \cos x = \frac{1}{\sqrt{2}}$

$y = \cos x \qquad m_2 = y' = -\sin x = -\frac{1}{\sqrt{2}}$

$\sin x = \cos x$

$\tan x = 1$

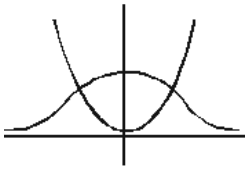
$x=\frac{\pi}{4},\frac{5\pi}{4}$

$$\tan\theta=\left|\frac{M_1-M_2}{1+M_1M_2}\right|=\left|\frac{\sqrt{2}}{1-\frac{1}{2}}\right|$$

$\theta = \tan^{-1}2\sqrt{2}$

(II) $x^2 = 4y \qquad y=\frac{8}{x^2+4}$

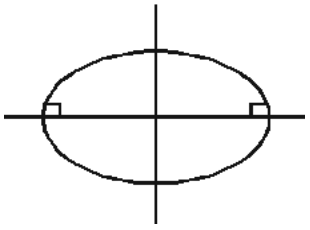
POI $\qquad y=\frac{8}{4y+4}$



$y^2 + y = 8$

$y=\frac{-1\pm\sqrt{1+32}}{2} \qquad 2 = n$

(III) $\frac{x^2}{18}+\frac{y^2}{8}=1$ and $y = 0$



$n = 2$

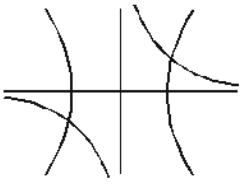
$\theta=\frac{\pi}{2}$

(IV) $xy = 1 \qquad x^2 - y^2 = 5$

$xy' + y = 0 \qquad 2x - 2yy' = 0$

$m_1 = y' = -y/x \quad m_2 = y' = \frac{x}{y}$

$m_1m_2 = -1 \qquad \theta=\frac{\pi}{2}$



$n = 2$

2. Ans (C)

(I) $y = \sin x \qquad m_1 = y' = \cos x = \frac{1}{\sqrt{2}}$

$y = \cos x \qquad m_2 = y' = -\sin x = -\frac{1}{\sqrt{2}}$

$\sin x = \cos x$

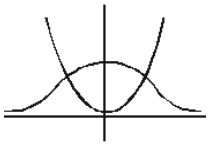
$\tan x = 1$

$x = \frac{\pi}{4}, \frac{5\pi}{4}$

$\tan\theta = \left|\frac{M_1 - M_2}{1 + M_1 M_2}\right| = \left|\frac{\sqrt{2}}{1 - \frac{1}{2}}\right|$

$\theta = \tan^{-1} 2\sqrt{2}$

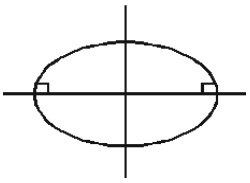
(II) $x^2 = 4y \qquad y = \frac{8}{x^2 + 4}$

POI $\quad y = \frac{8}{4y + 4}$ 

$y^2 + y = 8$

$y = \frac{-1 \pm \sqrt{1 + 32}}{2} \qquad 2 = n$

(III) $\frac{x^2}{18} + \frac{y^2}{8} = 1$ and $y = 0$

 $n = 2$

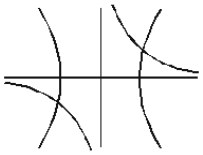
$\theta = \frac{\pi}{2}$

(IV) $xy = 1 \qquad x^2 - y^2 = 5$

$xy' + y = 0 \qquad 2x - 2yy' = 0$

$m_1 = y' = -y/x \qquad m_2 = y' = \frac{x}{y}$

$m_1 m_2 = -1 \qquad \theta = \frac{\pi}{2}$



$n = 2$

3. Ans (D)

(I) $y = \sin x \qquad m_1 = y' = \cos x = \frac{1}{\sqrt{2}}$

$y = \cos x \qquad m_2 = y' = -\sin x = -\frac{1}{\sqrt{2}}$

$\sin x = \cos x$

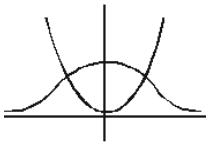
$\tan x = 1$

$x = \frac{\pi}{4}, \frac{5\pi}{4}$

$\tan\theta = \left|\frac{M_1 - M_2}{1 + M_1 M_2}\right| = \left|\frac{\sqrt{2}}{1 - \frac{1}{2}}\right|$

$\theta = \tan^{-1} 2\sqrt{2}$

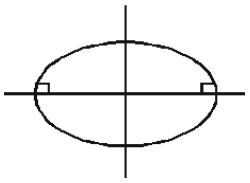
(II) $x^2 = 4y \qquad y = \frac{8}{x^2 + 4}$

POI $\quad y = \frac{8}{4y + 4}$ 

$y^2 + y = 8$

$y = \frac{-1 \pm \sqrt{1 + 32}}{2} \qquad 2 = n$

(III) $\frac{x^2}{18} + \frac{y^2}{8} = 1$ and $y = 0$

 $n = 2$

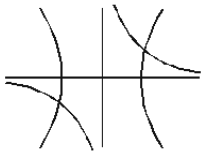
$\theta = \frac{\pi}{2}$

(IV) $xy = 1 \qquad x^2 - y^2 = 5$

$xy' + y = 0 \qquad 2x - 2yy' = 0$

$m_1 = y' = -y/x \qquad m_2 = y' = \frac{x}{y}$

$m_1 m_2 = -1 \qquad \theta = \frac{\pi}{2}$



$n = 2$

4. Ans (C)

(I) $y = \sin x + \cos x$ **(II)** $y = \sin 2x$

$y' = \cos x - \sin x$ $y' = 2\cos 2x$

$y \uparrow \left(0, \frac{\pi}{4}\right)$ $y \uparrow \left(0, \frac{\pi}{4}\right)$

$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ $y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(III) $y = \sin^5 x + \cos^5 x - 1$

$y' = 5\sin^4 x \cos x - 5\cos^4 x \cos x$

$= 5\sin x \cos x (\sin^3 x - \cos^3 x)$

$y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{\pi}{4}\right)$

$y = x^e$ $y' = x^x(1 + \ell n x)$

$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{1}{e}\right)$

5. Ans (D)

(I) $y = \sin x + \cos x$ **(II)** $y = \sin 2x$

$y' = \cos x - \sin x$ $y' = 2\cos 2x$

$y \uparrow \left(0, \frac{\pi}{4}\right)$ $y \uparrow \left(0, \frac{\pi}{4}\right)$

$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ $y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(III) $y = \sin^5 x + \cos^5 x - 1$

$y' = 5\sin^4 x \cos x - 5\cos^4 x \cos x$

$= 5\sin x \cos x (\sin^3 x - \cos^3 x)$

$y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{\pi}{4}\right)$

$y = x^e$ $y' = x^x(1 + \ell n x)$

$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{1}{e}\right)$

6. Ans (B)

(I) $y = \sin x + \cos x$ **(II)** $y = \sin 2x$

$y' = \cos x - \sin x$ $y' = 2\cos 2x$

$y \uparrow \left(0, \frac{\pi}{4}\right)$ $y \uparrow \left(0, \frac{\pi}{4}\right)$

$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ $y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(III) $y = \sin^5 x + \cos^5 x - 1$

$y' = 5\sin^4 x \cos x - 5\cos^4 x \cos x$

$= 5\sin x \cos x (\sin^3 x - \cos^3 x)$

$y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{\pi}{4}\right)$

$y = x^e$ $y' = x^x(1 + \ell n x)$

$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{1}{e}\right)$

MATHEMATICS

SEC-3

1. Ans (231)

(P) $L = \lim_{T\to\infty} \frac{1}{T} \int_0^{\frac{T}{2\pi}2\pi} (\sin x + \sin ax)^2 dx$

$= \lim_{T\to\infty} \frac{1}{2\pi} \int_0^{2\pi} (\sin x + \sin ax)^2 dx$

$a = 0 \quad L = \frac{1}{2\pi} \int_0^{2\pi} (\sin x + 0)^2 dx$

$= \frac{1}{2\pi} \int_0^{2\pi} \frac{1 + \cos 2x}{2} dx$

$= \frac{1}{4\pi} 2\pi = \frac{1}{2}$

$a = 1 \quad L = \lim_{T\to\infty} \frac{1}{2\pi} \int_0^{2\pi} (2\sin x)^2 dx$

$= \frac{4}{2\pi} \frac{1}{2} 2\pi = 2$

$a = -1 \quad L = 0$

2. Ans (213)

(P) $I = \int_{-1}^{1} \frac{1+x^3-\sqrt{1+x^6}}{1+x^6+2x^3-1-x^6}$

$= \int_{-1}^{1} \frac{1}{x^3} + \frac{1}{2}\int_{-1}^{1} dx - \int_{-1}^{1} \frac{\sqrt{1+x^6}}{2x^3}$

$= 1$

(Q) $f(x) = \frac{1}{x} - \frac{1}{x^2}$

$f'(x) = -\frac{1}{x^2} + \frac{2}{x^3}$

$f''(x) = +\frac{2}{x^3} - \frac{6}{x^4} = 0$

$x = 3$

(R) $I = \int_{1}^{e} \frac{1}{x(\ell nx+1)}$

$\ell nx = t$

$\frac{1}{x}dx = dt$

$= \int_{0}^{1} \frac{dt}{t+1} = (\ell n/t+1)_0^1$

$= \ell n2$

3. Ans (132)

(P) expand

(Q) $|C| = |B|\,|A|\,|B|^{-1}|B|^{-1}\,|A|\,|B|$

$= |A|^2 = 4^2 = 16$

(R) $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = I + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = I + B$

$B^2 = I$

$A^4 = (I+B)^4$

$= {}^4C_0I + {}^4C_1B + {}^4C_2B^2 + {}^4C_3B^3 + {}^4C_4B^4$

$= -4I$

4. Ans (221)

(P) $I = \int_{0}^{1} \ell n\frac{1-x}{x}$ Property (5) and add I = 0

(Q)

$\int_{0}^{10\pi} \lim_{x\to y}\left(\frac{\sin x - \sin y}{x-y}\right)dx = \int_{0}^{10\pi} \cos y\,dy$

$= 0$

(R) $I = \sum \frac{1^2}{1^3+n^3} = \int_{0}^{2} \frac{x^2}{x^3+1}dx$

$x^3 + 1 = t$

$= \int_{1}^{9} \frac{\frac{1}{3}dt}{t} = \frac{1}{3}(\ell nt)_1^9$

$= \frac{1}{3}\ell n9$

5. Ans (122)

(P) $\sqrt{x^2+y^2}=\dfrac{y\sqrt{1+m^2}}{m}$

$x^2m^2+m^2y^2=y^2+m^2y^2$

$\dfrac{dy}{dx}=\pm\dfrac{y}{x}$

$\int\dfrac{dy}{y}=\pm\int\dfrac{dx}{x}$

$\ell ny=\pm\ell nx+\ell nc$

$xy=k$ $\quad$ y = k'x straight line

(Q) tangent $y-g=m(x-x)$

B(0,y – mx) $\quad$ P(x,y)

PB = OB

$\sqrt{x^2+m^2x^2}=|y-mx|$

$x^2+m^2x^2=y^2+m^2x^2-zmxy$

$2xy\dfrac{dy}{dx}=g^2-x^2$

$\dfrac{2xydy-y^2dx^2}{x^2}=\dfrac{-x^2dx}{x^2}$

$d\left(\dfrac{y^2}{x}\right)=-\int dx$

$\dfrac{y^2}{x}=-x+c$

$y^2+x^2+cx=0$ $\quad$ circle

(R) $\sqrt{x^2+y^2}=y_1\sqrt{1+m^2}$

$x^2+y^2=y^2+y^2m^2$

$\dfrac{dy}{dx}=\pm\dfrac{x}{y}$

$\dfrac{y^2}{2}=\pm\dfrac{x^2}{2}+c$

circle

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
27-06-2021

JEE(Main + Advance) : IIT-JEE(M+A) XII & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 27-06-2021

ANSWER KEY PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,C	C,D	A,C	A,D	B,D	A,B	A,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	C	D	C	B	A	
SEC-3	Q.	1	2	3	4	5		
	A.	112	213	321	312	213		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,D	A,B,D	B,D	A,B,C,D	A,B,C,D	A,B,D	B,C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	D	D	A	D	D	
SEC-3	Q.	1	2	3	4	5		
	A.	312	321	321	213	231		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	B,C	A,D	B,C	C,D	A,B,C	C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	A	C	D	C	D	B	
SEC-3	Q.	1	2	3	4	5		
	A.	231	213	132	221	122		

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Main)
MAJOR
27-06-2021

JEE(Main) : IIT-JEE(M+A) XII & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Main)

TEST DATE : 27-06-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	D	A	B	C	C	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	B	B	A	A	B	A	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.60	6.25	75.00	7.00	14.40	2.50	10.00	1.00	0.00	312.50

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	A	B	A	A	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	A	C	B	C	B	A	D	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	120.00	19.00	11.00	700.00	-38.00	2.00	202.00	16.00	4.09	500.00

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	B	C	A	D	D	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	A	D	A	D	A	B	B	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6.00	2.00	4.00	5.33	50.00	1.50	2.75	2.50	1.41, 1.42	3.00

HINT – SHEET

PHYSICS

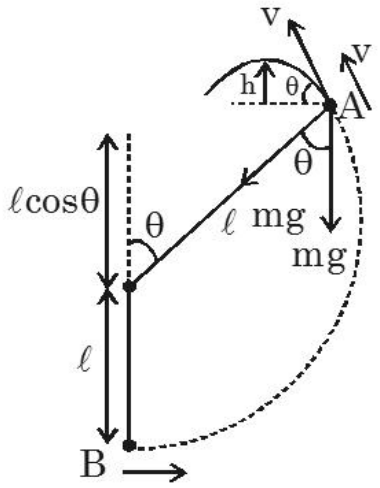
SEC-A

1. Ans (C)

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

n_1, n_2 can not be greater than 4

2. Ans (D)



From B to A

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = -mg(\ell + \ell\cos\theta)$$

$$v^2 - u^2 = -2g\ell(1+\cos\theta) \quad \text{... (i)}$$

$$mg\cos\theta = \frac{mv^2}{\ell}$$

$$v^2 = g\ell\cos\theta \quad \text{... (ii)}$$

From (i) & (ii)

$$g\ell\cos\theta - \frac{7}{2}g\ell = -2g\ell - 2g\ell\cos\theta$$

$$3g\ell\cos\theta = \frac{3}{2}g\ell$$

$$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$h = \frac{v^2\sin^2\theta}{2g} = \frac{g\ell\cos 60^\circ \sin^2 60^\circ}{2g}$$

$$h = \frac{\ell}{2}\left(\frac{1}{2}\right)\left(\frac{3}{4}\right) = \frac{3\ell}{16}$$

$$H = \ell + \ell\cos\theta + \frac{3\ell}{16} = \left(\frac{16+8+3}{16}\right)\ell$$

$$= \frac{27}{16}\ell$$

3. Ans (C)

(1) $r_n \propto \frac{n^2}{z}$

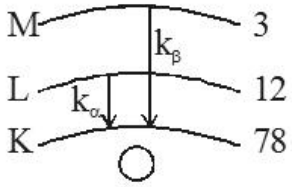
(2) $v_n = \frac{2\pi e^2}{h}\left(\frac{z}{n}\right)$

(3) v_n does not depend on mass of electron

(4) $T = \frac{2\pi r_n}{v_n}$

$$T \propto \frac{n^2/z}{z/n} \Rightarrow T \propto \frac{n^3}{z^2}$$

4. Ans (D)

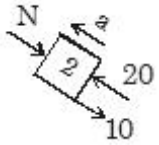


$$\frac{\left(\frac{hc}{\lambda_{k\alpha}}\right)}{\left(\frac{hc}{\lambda_{k\beta}}\right)} = \frac{(E_k - E_L)}{(E_k - E_m)}$$

$$\Rightarrow \frac{\lambda_{k\beta}}{\lambda_{k\alpha}} = \frac{78-12}{78-3} = \frac{22}{25} = \frac{\lambda_{k\alpha}}{\lambda_{k\beta}} = \frac{25}{22}$$

5. Ans (A)

$$a = \frac{20-2-15}{3} = 1\,m/s^2$$



$20 - 10 - N = 2 \times a$

$N = 10 - 2 = 8N$

6. Ans (B)

Use linear momentum conservation $\vec{P}_i = \vec{P}_f$

8. Ans (C)

Let initially they have n atoms

A → the half time two hours

B → the half time your hours

After two hours A posses 1 half life and B posses $\frac{1}{2}$ half life

Remaining atoms of A = $\frac{x}{2^1}$

Remaining atoms of B = $\frac{n}{(2)^{\frac{1}{2}}}$

9. Ans (D)

$a_x = 0 \Rightarrow \Delta v_x = 0$

$a_y = -g \Rightarrow \Delta v_y = -g\Delta t$

10. Ans (A)

Area of F-t graph = Impulse

Now using impulse momentum theorem

$\int F\,dt = \Delta P$

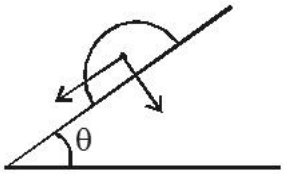
We get = $v = 2.9 \times 10^{-3}$ m/s.

11. Ans (D)

$a_{cm} = \frac{F - \mu Mg}{M}$, $F\frac{\ell}{3} - \mu Mg\frac{\ell}{2} = \frac{M\ell^2}{12}\alpha$

$a_A = a_{cm} - \frac{\ell}{2}\alpha = 2\mu g - \frac{F}{M}$

12. Ans (B)



$mg\cos\theta \times R = mg\sin\theta \times \frac{2R}{\pi}$

$\Rightarrow \tan\theta = \frac{\pi}{2}$

13. Ans (C)

$v_{cm} = \frac{20 \times 6 - 10 \times 3}{20 + 10} = 3$ m/s towards right.

At the time of either maximum elongation or compression, both the blocks will be moving with a common velocity = 3 m/s towards right.

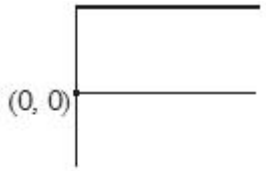
14. Ans (B)

For a compound pendulum

$T = 2\pi\sqrt{\frac{I_S}{mgd}}$

$T = 2\pi\sqrt{\frac{\frac{10}{3}\frac{m}{4}\ell^2\sqrt{2}}{mg \times \ell}} = 2\pi\sqrt{\frac{5\sqrt{2}\ell}{6g}}$

15. Ans (B)



$x_{cm} = \frac{mx_0 + 2m \times \ell/2}{3m} = \frac{\ell}{3}$

$y_{cm} = \frac{2mx_0 + m \times \ell/2}{3m} = \frac{\ell}{6}$

$COM = \left(\frac{\ell}{3}, \frac{\ell}{6}\right)$

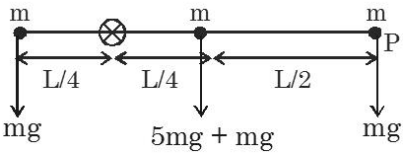
16. Ans (A)

$W_f = -\mu mgx$

$= -0.3 \times 2 \times 10 \times 5$

$= -30$ J

17. Ans (A)



$\tau_0 = I_0\alpha$

$6mg\frac{L}{4} + mg\frac{3L}{4} - mg\frac{L}{4}$

$= \left(\frac{5mL^2}{12} + \frac{5mL^2}{16} + \frac{mL^2}{16} \times 2 + \frac{9mL^2}{16}\right)\alpha$

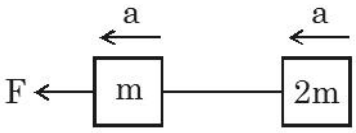
$2g = \left(\frac{17}{12}L\right)\alpha$

$\alpha = \frac{24g}{17L}$

$a_P = \frac{3}{4}L\alpha = \frac{3}{4}L \times \frac{24g}{17L}$

$a_P = \frac{18}{17}g$

18. Ans (B)



$F = 3ma$

$a = \frac{F}{3m}$

19. Ans (A)

Use impulse momentum theorem.

Impulse = Change in momentum

$\vec{J} = \Delta\vec{P}$

20. Ans (A)

Since 10 lines are emitted hence e– is in n = 5 state.

Maximum wavelength is corresponding to transition n = 5 to n = 4

$\frac{1}{\lambda} = R\left[\frac{1}{16} - \frac{1}{25}\right]$

$\lambda = 4.05 \times 10^{-6}$ m

PHYSICS

SEC-B

1. Ans (1.60)

$E = \frac{hc}{\lambda} = \frac{12400}{0.2 \times 10^4} = 6.2eV$

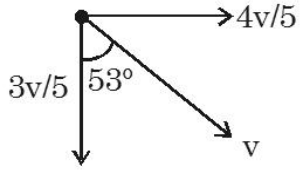
$KE_{max} = E - \phi = 6.2 - 4.6 = 1.6$ eV

$eV_0 = 1.6$ eV

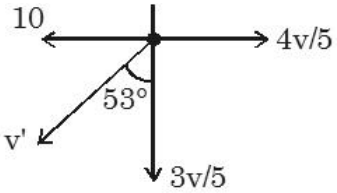
$V_0 = 1.6$ Volts

2. Ans (6.25)

1st Case



2nd Case,



$10 - \frac{4v}{5} = \frac{4v}{5}$

$\frac{8v}{5} = 10$

$v = \frac{25}{4}$ m/s

3. Ans (75.00)

$R = \frac{2u^2 \cos\alpha \sin(\alpha - \beta)}{g\cos^2\beta}$

Putting

$\alpha = 53°$ & $\alpha = 37°$ we get

R = 75m

4. Ans (7.00)

$\frac{1}{\lambda} = R(13-1)^2\frac{3}{4}$

$\frac{1}{4\lambda} = R(Z-1)^2\frac{3}{4}$

Solving we get Z = 7

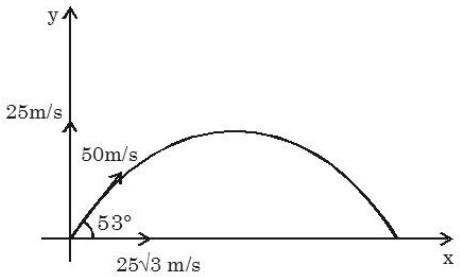
5. Ans (14.40)

$T_{1/2} = \frac{36 \times 24}{36 + 24}$

6. Ans (2.50)

$2a = v\frac{dm}{dt} = 5 \times 1$

7. Ans (10.00)



$\vec{u} = 25\sqrt{3}\,\hat{i} + 25\,\hat{j}$:

$\vec{v} = 25\sqrt{3}\,\hat{i} + (25 - gt)\,\hat{j}$

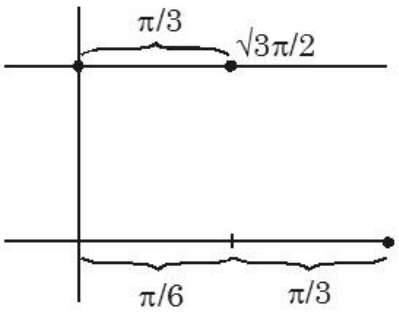
$\vec{u} \cdot \vec{v} = 0$

$25\sqrt{3} \times 25\sqrt{3} + 25\,(25 - 10t) = 0$

$25 - 10t = -75$

$10t = 100 \Rightarrow t = 10$ sec.

8. Ans (1.00)



(i) $t_1 = \frac{\pi}{3} \times \frac{T}{2\pi} = \frac{T}{6}$

(ii) $t_2 = \frac{\pi}{3} \times \frac{T}{2\pi} = \frac{T}{6}$

Ratio = $\frac{T/6}{T/6} = 1$

9. Ans (0.00)

Since no relative motion, accelerations of both blocks will be same.

$20 - f = 4a$

$10 + f = 2a$

Solving we get $f = 0$

10. Ans (312.50)

At t = 7 sec

$v_x = u_x = 40$ m/s

$v_y = 40 - 10 \times 7 = -30$ m/s

angle made by velocity vector with horizontal

$\tan\alpha = \frac{v_y}{v_x} = \frac{3}{4}$

$R = \frac{v^2}{a_\perp} = \frac{v^2}{g\cos\alpha} = \frac{50 \times 50}{10 \times 4} \times 5$

$R = \frac{625}{2}$ m

CHEMISTRY

SEC-A

1. Ans (A)

In adiabatic free expansion temperature does not change.

2. Ans (B)

Correct order of melting point is :

Ti < V < Cr > Mn

4. Ans (A)

If $T_1 = T_2 = T_3$ then molecular mass of gas 'A' > molecular mass of gas 'B' > molecular mass of gas 'C'.

5. Ans (B)

SiH_4 is electron precise molecular hydride.

6. Ans (A)

In Calgon process water soluble complex is formed with $Na_6P_6O_{18}$.

10. Ans (A)

Misch metal is ~95% Lanthanoides and ~5% Fe

15. Ans (B)

Solution of alkali metal in liquid NH3 shows reducing character.

17. Ans (B)

$$2LiNO_3 \longrightarrow Li_2O + 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$$

In KO_2, superoxide ion is present which is paramagnetic in nature.

CHEMISTRY

SEC-B

1. Ans (120.00)

$\Delta H = \Delta U + \Delta(PV)$

$= \Delta U + (P_2V_2 - P_1V_1)$

$= 50kJ + (80\times15 - 50\times10) \times 100$ Joules

$= 50$ kJ + 70 kJ

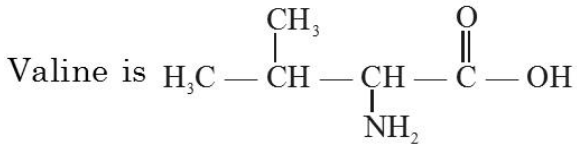
$= 120$ kJ

2. Ans (19.00)

In acidic medium $KMnO_4$ undergoes redox reaction with $H_2C_2O_4$, $Na_2S_2O_3$ and HNO_2 and converted into Mn^{2+}.

Total sum of atoms = 8 + 7 + 4 = 19

3. Ans (11.00)

Valine is 

4. Ans (700.00)

$q_A = q_B$

$M_1S_1\Delta T = M_2S_1\Delta T$

$50 \times 1(T - 46) = 70 \times 1 \times (70 - T)$

$5T - 230 = 490 - 7T$

$12T = 720$

$T = 60°$

$\Delta H_A = q = 50\times14 \Rightarrow 700$ calories

6. Ans (2.00)

In reaction (2) H_2O_2 reduces MnO_2 to $MnSO_4$ in acidic medium. So act as reducing agent.

n-factor (MnO_2) = 2

7. Ans (202.00)

$V_{mp} = 100$ m/sec

$$V_{avg} = \frac{(60\times10)+(80\times20)+(100\times40)+(120\times15)+(140\times10)+(160\times5)}{10+20+40+15+10+5}$$

= 102 m/sec

$V_{mp} + V_{avg} = 202$ m/sec

9. **Ans (4.09)**

$$\text{Ratio is } \frac{303+303}{75+73} = 4.09$$

10. **Ans (500.00)**

$\Delta H^\circ_{\text{reaction}} = -375 - (-290)$

$= -85$ kJ/mole

$\Delta S^\circ = 360 - 220 - 310$

$= -170$ J/mole/K

For equilibrium $\Delta G^\circ = 0$

$\Delta H^\circ = T\Delta S^\circ$

$$T = \frac{\Delta H^\circ}{\Delta S^\circ}$$

$$= \frac{-85}{-170} \times 1000 = 500\text{K}$$

MATHEMATICS

SEC-A

1. **Ans (A)**

$$I = 5\int_0^{2\pi} \sqrt{2}\sin\left(x+\frac{\pi}{4}\right)dx$$

$$I = 5\left[\int_0^{\pi/2} 1\,dx + \int_{3\pi/4}^{\pi} (-1)\,dx + \int_{\pi}^{3\pi/2} (-2)\,dx + \int_{3\pi/2}^{7\pi/4} (-1)\,dx\right]$$

$= -5\pi$

2. **Ans (C)**

$f(x) + f(x+1) + f(x+2) + f(x+3) = 2021$

$\Rightarrow f(x+1) + f(x+2) + f(x+3) + f(x+4) = 2021$

$\Rightarrow f(x) - f(x+4) = 0$

$\Rightarrow f(x) = f(x+4)$

$$\text{Now } \int_0^{100} f(x)\,dx = 25\int_0^4 f(x)\,dx$$

$= 25A$

3. **Ans (D)**

$f'(x) = (6(g(x))^2 - 6g(x) + 3)\,g'(x) - 2 - \sin x$

if $g'(x) < 0 \Rightarrow f'(x) < 0$

4. **Ans (B)**

Required angle will be the angle between $3x + 4y = 0$ and $4x - 3y = 0$ is 90°.

5. **Ans (C)**

$$y' = \frac{8x}{3(4-y^2)} \Rightarrow x = 0$$

$\Rightarrow y = 0, \pm 2\sqrt{3}$

6. **Ans (A)**

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{h}{1+|h|}}{h} = 1$$

$$\frac{dy}{dx} = \begin{cases} \frac{1}{(1-x)^2}, & x < 0 \\ \frac{1}{(1+x)^2}, & x > 0 \end{cases}$$

Required tangent $y = x$

7. Ans (D)

$A = A^T$

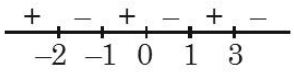
as $|A| = -1 \Rightarrow A^T = -\text{adj}A$

$\Rightarrow A^2 = I \Rightarrow A^4 = A^2$

8. Ans (D)

$f'(x) = (e^{-x} - 1)(x^2 - 1)^5 (x + 2)^9 (x - 3)^7$

minima at 1 and –1



9. Ans (C)

$$\int_0^{10} f(x)\,dx = \int_0^5 f(x)\,dx + \int_5^{10} f(x)\,dx$$

$$= 7 + \int_5^{10} f(x)\,dx$$

Let $t = 10 - x$

$$7 + \int_5^0 -f(10 - t)\,dt$$

$= 7 + 7 = 14$

10. Ans (A)

$$\int (2\sin(2\ell nx) + \sin^2 \ell nx - 1)\,dx$$

$$= \int (4\sin(\ell nx)\cos(\ell nx) + 2\sin^2 \ell nx - 1)\,dx$$

$f(x) = 2\sin^2 \ell nx$

$$f'(x) = \frac{4\sin(\ell nx)\cos(\ell nx)}{x}$$

$$= \int (xf'(x) + f(x))\,dx - x$$

$= 2x\sin^2(\ell nx) - x + c$

$= 2e^{\pi/2} - e^{\pi/2} + c = 0$

$\Rightarrow c = -e^{\pi/2}$

11. Ans (D)

$$\text{subnormal} = y\frac{dy}{dx}$$

$$ny^{n-1}\frac{dy}{dx} = \frac{-k^{n+1}}{x^2} \Rightarrow \frac{dy}{dx} = \frac{-k^{n+1}}{nx^2y^{n-1}}$$

$$\Rightarrow y\frac{dy}{dx} = \frac{-k^{n+1}}{nx^2y^{n-2}} = \frac{-k^{n+1}y^{2n}}{n(k^{n+1})^2y^{n-2}}$$

$$= \frac{-y^{n+2}}{nk^{n+1}}, \text{ for constant , } n = -2$$

12. Ans (C)

$XY = AB^TBA^T = I$

$\Rightarrow \text{tr}(XY) = 3$

13. Ans (A)

$f'(x) = 4x^3 + 18x^2 + 70x + 6$

and $f''(x) = 12x^2 + 36x + 70 > 0 \quad \forall\, x \in R$

so $f(x)$ has no point of inflection.

14. Ans (D)

$$\int \frac{xdy + ydx}{\sqrt{1 - (xy)^2}} = \int dx$$

$$\Rightarrow \sin^{-1}(xy) = x + c \Rightarrow xy = \sin(x + c)$$

15. Ans (A)

$xy(xdx + ydy) = xdy + ydx$

$$\Rightarrow xdx + ydy = \frac{xdy + ydx}{xy}$$

on integrating

$$\frac{x^2 + y^2 + c}{2} = \ell n(xy)$$

$$\Rightarrow x^2y^2 = e^{x^2+y^2+c}$$

16. Ans (D)

$f'(x) + x^2 f(x) + 2x = 0$

$f'(1) + 1 + 2 = 0$

$\Rightarrow f'(1) = -3$

17. Ans (A)

$f(x)$ is an even function

$$\therefore \frac{dy}{dx} = 0 \text{ at } (0,0)$$

$\Rightarrow$ point of intersection with y-axis is (0,–1)

$\Rightarrow$ equation of tangent is $y = -1$

18. Ans (B)

For unique solution

$$\begin{vmatrix} 1 & -1 & 1 \\ \lambda & 3 & 2 \\ 3 & \lambda & 1 \end{vmatrix} \neq 0$$

$\Rightarrow \lambda \neq 4, -3$

19. Ans (B)

$|\text{adj}\lambda A| = |\lambda A|^2 = \lambda^6 |A|^2$

$= \lambda^6.6^2 = a^b.b^a$

$\Rightarrow \lambda = 2, a = 6$

20. Ans (A)

$|A| = -1$

$\det(A^{2019} - 3A^{2018}) = |A^{2018}(A - 3I)|$

$= (-1)^{2018}\begin{vmatrix} -1 & 7 \\ 1 & 0 \end{vmatrix} = -7$

MATHEMATICS

SEC-B

1. Ans (6.00)

$$\int e^x \frac{\tan\left(x - \frac{\pi}{4}\right)}{2} + \frac{1}{1 - \cos\left(\frac{\pi}{2} - 2x\right)} dx$$

$$= \int e^x \frac{\left(\tan\left(x - \frac{\pi}{4}\right)\right)}{2} + \frac{\sec^2\left(x - \frac{\pi}{4}\right)}{2} dx$$

$$= \frac{1}{2} e^x \tan\left(x - \frac{\pi}{4}\right) + c$$

$$\Rightarrow f(x) = \frac{\tan\left(x - \frac{\pi}{4}\right)}{2}$$

2. Ans (2.00)

$f'(x) = 0 \Rightarrow f(x) = k$

3. Ans (4.00)

Let $h(x) = f(x) - g(x)$

$\Rightarrow h(x) = 0$ has exactly 7 distinct roots

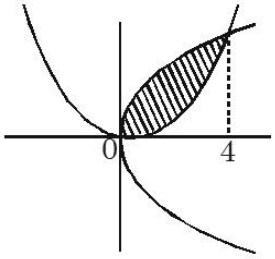
$\Rightarrow h'(x) = 0$ has at least 6 roots

$\Rightarrow h''(x) = 0$ has at least 5 roots

$\Rightarrow h'''(x) = 0$ has at least 4 roots

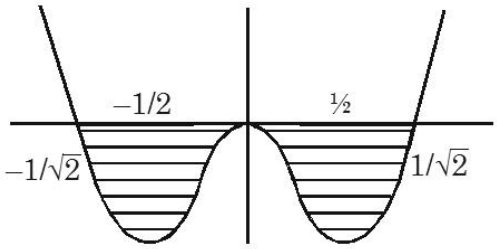
$h'''(x) = f'''(x) - 12$

4. Ans (5.33)



$$\int_0^4 \left(\sqrt{4x} - \frac{x^2}{4}\right) dx = \frac{16}{3}$$

5. Ans (50.00)



Required area $= 2\int_0^{1/2} \left|2x^4 - x^2\right| dx = \frac{7}{120}$

6. Ans (1.50)

$f'(x) = 0 \Rightarrow x = -2$ & 3

leading co-efficient of $f(x)$ is positive

So, $f(x)$ has maximum at $x = -2$ and minimum at $x = 3$

$\Rightarrow \left|\frac{q}{p}\right| = \frac{3}{2} = 1.50$

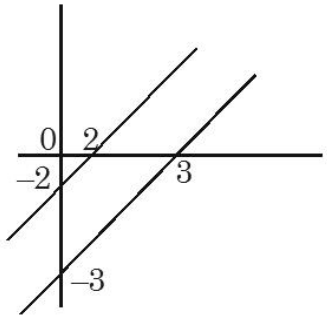
7. Ans (2.75)

Required Area $= \left|\int_{\pi/4}^{\pi/3} f(x) - g(x)\right|$

$$= \int_{\pi/4}^{\pi/3} \left(\cos 2x + \sec^2 x - \operatorname{cosec}^2 x\right) dx$$

$$= \left(\frac{\sin 2x}{2} + \tan x + \cot x\right)_{\frac{\pi}{4}}^{\frac{\pi}{3}} = \frac{19\sqrt{3} - 30}{12}$$

8. Ans (2.50)



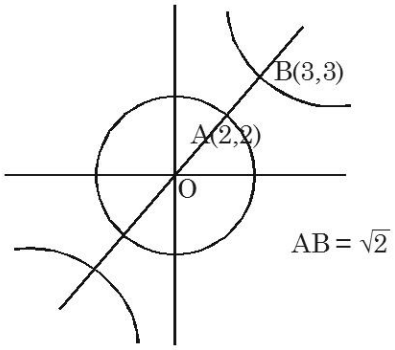
$(x-y)^2-5(x-y)+6=0$

$\Rightarrow x-y=2$ and $x-y=0$

$\text{Area}=\frac{1}{2}\times 3^2-\frac{1}{2}\times 2^2=\frac{5}{2}$

9. Ans (1.41, 1.42)

Along common normal



10. Ans (3.00)

$f(x)\geq 0\ \ \forall\ x\in R$

$3x^2+6ax+11a-6\geq 0\ \forall\ x\in R$

$\therefore D\leq 0\Rightarrow a\in\left[\frac{2}{3},3\right]$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Main)
MAJOR
27-06-2021

JEE(Main) : IIT-JEE(M+A) XII & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR

Test Pattern : JEE (Main)

TEST DATE : 27-06-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	D	A	B	C	C	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	B	B	A	A	B	A	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.60	6.25	75.00	7.00	14.40	2.50	10.00	1.00	0.00	312.50

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	A	B	A	A	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	A	C	B	C	B	A	D	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	120.00	19.00	11.00	700.00	-38.00	2.00	202.00	16.00	4.09	500.00

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	B	C	A	D	D	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	A	D	A	D	A	B	B	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6.00	2.00	4.00	5.33	50.00	1.50	2.75	2.50	1.41, 1.42	3.00